MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE KHQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.616 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KHQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/72-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the currentversion before use.

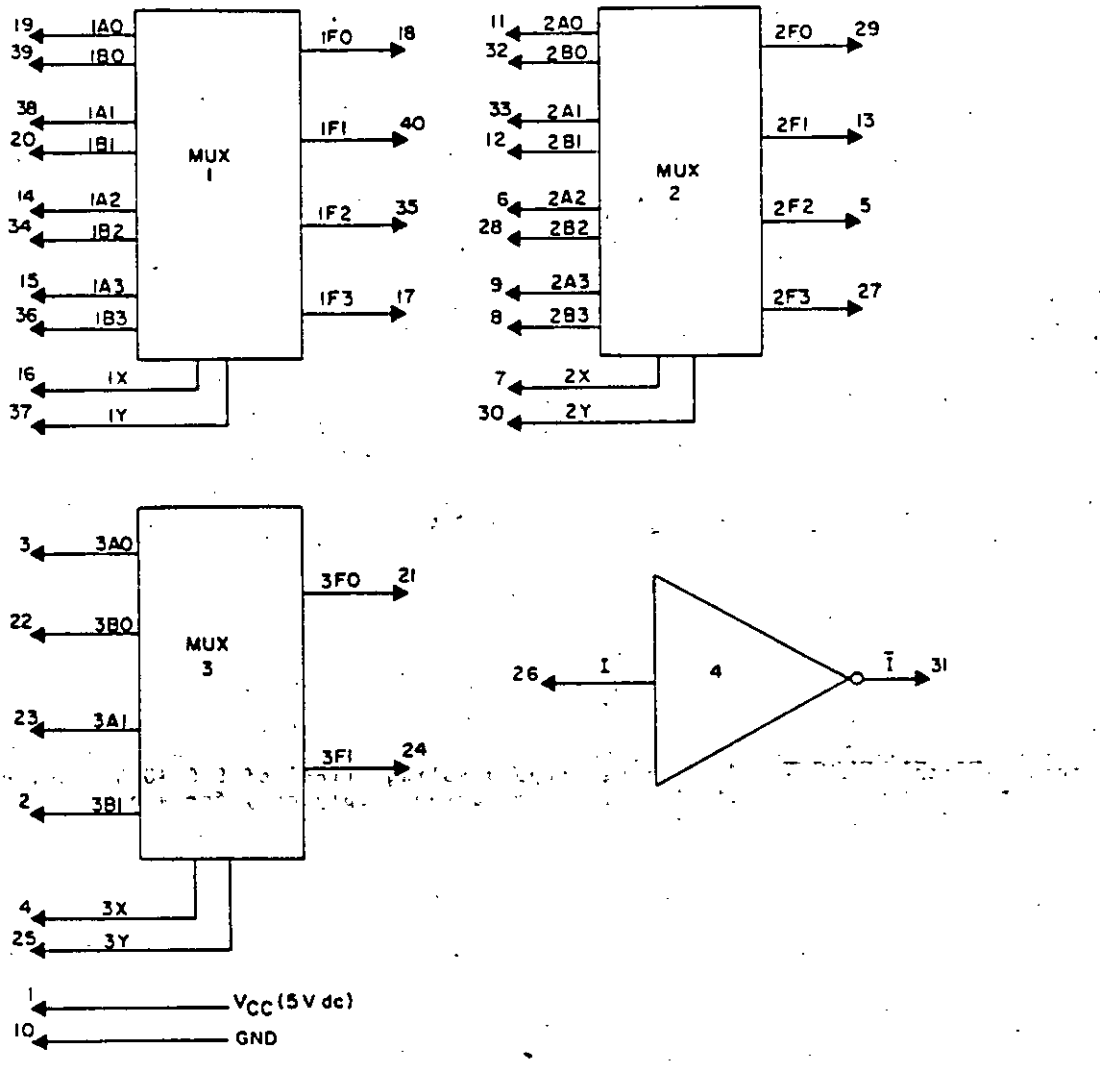


FIGURE 1. KHQ module function diagram, multiplexer.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	10	∞	10	∞	10	∞	MΩ
B1-B11	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C13	Input breakdown current	V	S	0	100	0	100	0	100	0	100	µA
C14-C27	Input breakdown current	V	S	0	200	0	200	0	200	0	200	µA
D1-D11	Output short circuit current	VI	X	-6	-135	-6	-135	-6	-135	-6	-135	mA
E1-E11	Output leakage current	VII	X	0	250	0	250	0	250	0	250	µA
F1-F11	Output voltage, high level	VIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
F12-F22	Output voltage, low level	VIII	X	0	500	0	500	0	500	0	500	mV
G1-G32	Functional	IX	X	Test vectors must produce output vectors per table IX								
H1-H27	Input clamp diode voltage	X	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
J1-J13	Input current, high level	XI	X	0	20	0	20	0	20	0	20	µA
J14-J27	Input current, high level	XI	X	0	40	0	40	0	40	0	40	µA
K1-K13	Input current, low level	XII	X	-0.10	-0.44	-0.10	-0.44	-0.10	-0.44	-0.10	-0.44	mA
K14-K27	Input current, low level	XII	X	-0.20	-0.88	-0.20	-0.88	-0.20	-0.88	-0.20	-0.88	mA
L	Power supply current, low level outputs	XIII	X	10	64	10	64	10	64	10	64	mA
M1-M11	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	10	50	10	50	10	55	10	55	ns
M12-M22	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	20	80	20	80	20	90	20	90	ns
N1-N11	Transition time (t_{THL}), C_L = 50 pF	XIV	S	3	18	3	18	3	20	3	20	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260120T15:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N12-N22	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	6	40	6	40	6	40	6	40	ns
P1-P10	Propagation delay time (t_{PHL}), A0-A3 to output, C_L = 50 pF	XV	S	5	31	5	31	5	33	5	33	ns
P11-P20	Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 50 pF	XV	S	5	31	5	31	5	33	5	33	ns
P21-P30	Propagation delay time (t_{PHL}), B0-B3 to output (inverting), C_L = 50 pF	XV	S	5	46	5	46	5	48	5	48	ns
P31-P70	Propagation delay time (t_{PHL}), X and Y address to output, C_L = 50 pF	XV	S	5	43	5	43	5	45	5	45	ns
P71	Propagation delay time (t_{PHL}), INVERTER, C_L = 50 pF	XV	S	3	19	3	19	3	21	3	21	ns
P72-P81	Propagation delay time (t_{PHL}), A0-A3 to output, C_L = 150 pF	XV	Q	5	41	5	41	5	43	5	43	ns
P82-P91	Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 150 pF	XV	Q	5	41	5	41	5	43	5	43	ns
P92-P101	Propagation delay time (t_{PHL}), B0-B3 to output (inverting), C_L = 150 pF	XV	Q	5	56	5	56	5	58	5	58	ns
P102-P141	Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	XV	Q	10	53	10	53	10	55	10	55	ns
P142	Propagation delay time (t_{PHL}), INVERTER, C_L = 150 pF	XV	Q	4	29	4	29	4	31	4	31	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements - 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q10	Propagation delay time (t_{PLH}), A0-A3 to output, C_L = 50 pF	XV	S	5	20	5	20	5	22	5	22	ns
Q11-Q20	Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 50 pF	XV	S	5	20	5	20	5	22	5	22	ns
Q21-Q30	Propagation delay time (t_{PLH}), B0-B3 to output (inverting), C_L = 50 pF	XV	S	5	35	5	35	5	37	5	37	ns
Q31-Q70	Propagation delay time (t_{PLH}), X and Y address to output, C_L = 50 pF	XV	S	5	34	5	34	5	36	5	36	ns
Q71	Propagation delay time (t_{PLH}), INVERTER, C_L = 50 pF	XV	S	3	19	3	19	3	21	3	21	ns
Q72-Q81	Propagation delay time (t_{PLH}), A0-A3 to output C_L = 150 pF	XV	Q	5	25	5	25	5	27	5	27	ns
Q82-Q91	Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 150 pF	XV	Q	5	25	5	25	5	27	5	27	ns
Q92-Q101	Propagation delay time (t_{PLH}), B0-B3 to output (inverting), C_L = 150 pF	XV	Q	5	40	5	40	5	42	5	42	ns
Q102-Q141	Propagation delay time (t_{PLH}), X to Y address to output, C_L = 150 pF	XV	Q	10	39	10	39	10	41	10	41	ns
Q142	Propagation delay time (t_{PLH}), INVERTER C_L = 150 pF	XV	Q	4	29	4	29	4	31	4	31	ns

---

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10 20T15:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R1-R11	Output crosstalk, low level	XVI	Q	0	600	0	600	0	600	0	600	mV
S1-S11	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V1-V12	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF
V13	Input capacitance	XX	Q	2	20	2	20	2	20	2	20	pF
V14-V27	Input capacitance	XX	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/72-1 supersedes part number M28787/72. The two parts are interchangeable.

*6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	352 mW (outputs not loaded)	
Supply voltage	5 V dc ±10 percent	
Input voltage		
Low level	-0.5 V minimum	0.6 V maximum
High level	2.0 V minimum	5.5 V maximum

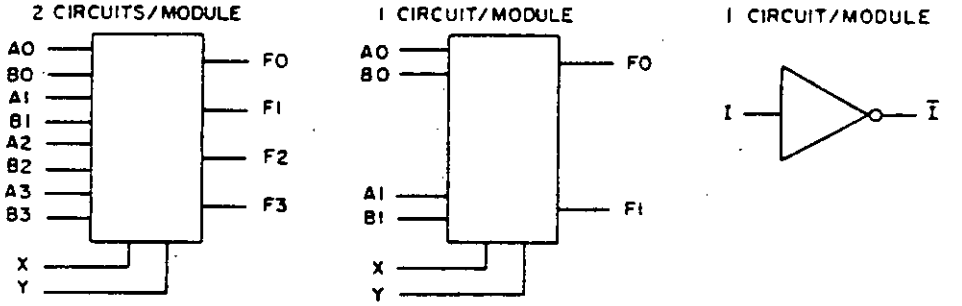
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

Input current	
Low level	
A, I, 3X, 3Y	
(V_{IL} = 0.0 V)	0.44 mA maximum
(V_{IL} = 0.4 V)	0.36 mA maximum
B, 1X, 2X, 1Y, 2Y	
(V_{IL} = 0.0 V)	0.88 mA maximum
(V_{IL} = 0.4 V)	0.72 mA maximum
High level	
A, I, 3X, 3Y	
(V_{IH} = 2.5 V)	20 µA maximum
(V_{IH} = 5.5 V)	100 µA maximum
B, 1X, 1Y, 2X, 2Y	
(V_{IH} = 2.5 V)	40 µA maximum
(V_{IH} = 5.5 V)	200 µA maximum
Output voltage	
Low level (I_{OL} = 6 mA)	0.5 V maximum
High level (I_{OH} = -400 µA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	6.0 mA
High level (V_{OH} = 2.5 V)	-400 µA

6.5 Module function. This module contains four circuits. Two circuits consist of four 2-way multiplexers with common select lines and one circuit consists of two 2-way multiplexers with common select lines. The module also contains one independent inverter circuit.

6.5.1 Input to output relationship. Each module has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.



X CONTROL	Y CONTROL	F OUTPUT
L	L	L
L	H	B
H	L	$\overline{B}$
H	H	A

I	$\overline{I}$
L	H
H	L

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 0 volt reference. The reference will be zero volt for the test setup and not necessarily earth ground.

6.7 Decoupling. Decoupling of test voltages and V_{CC} to minimize noise in the test setup is permissible except test condition T, power supply transient amplitude test.

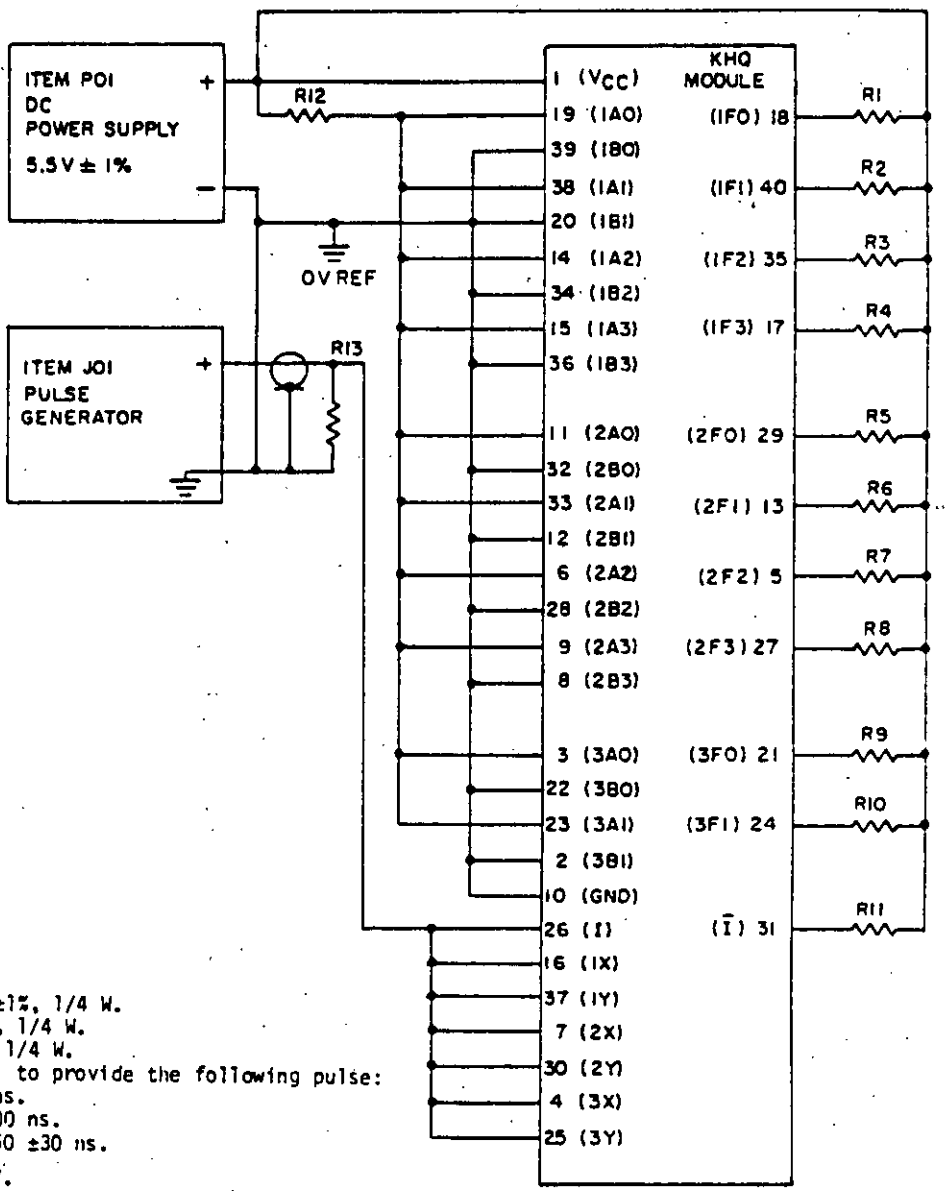
6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N445)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R11 = 845Ω ±1%, 1/4 W.
2. R12 = 1 kΩ ±5%, 1/4 W.
3. R13 = 51Ω ±5%, 1/4 W.
4. Adjust item J01 to provide the following pulse:
   PW = 500 ±100 ns.
   PRP = 1,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 50 ±30 ns.
   PA = 3.5 ±0.5 V.

* FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 20261Q-20T15:53Z
Check the source to verify that this is the current version before use.

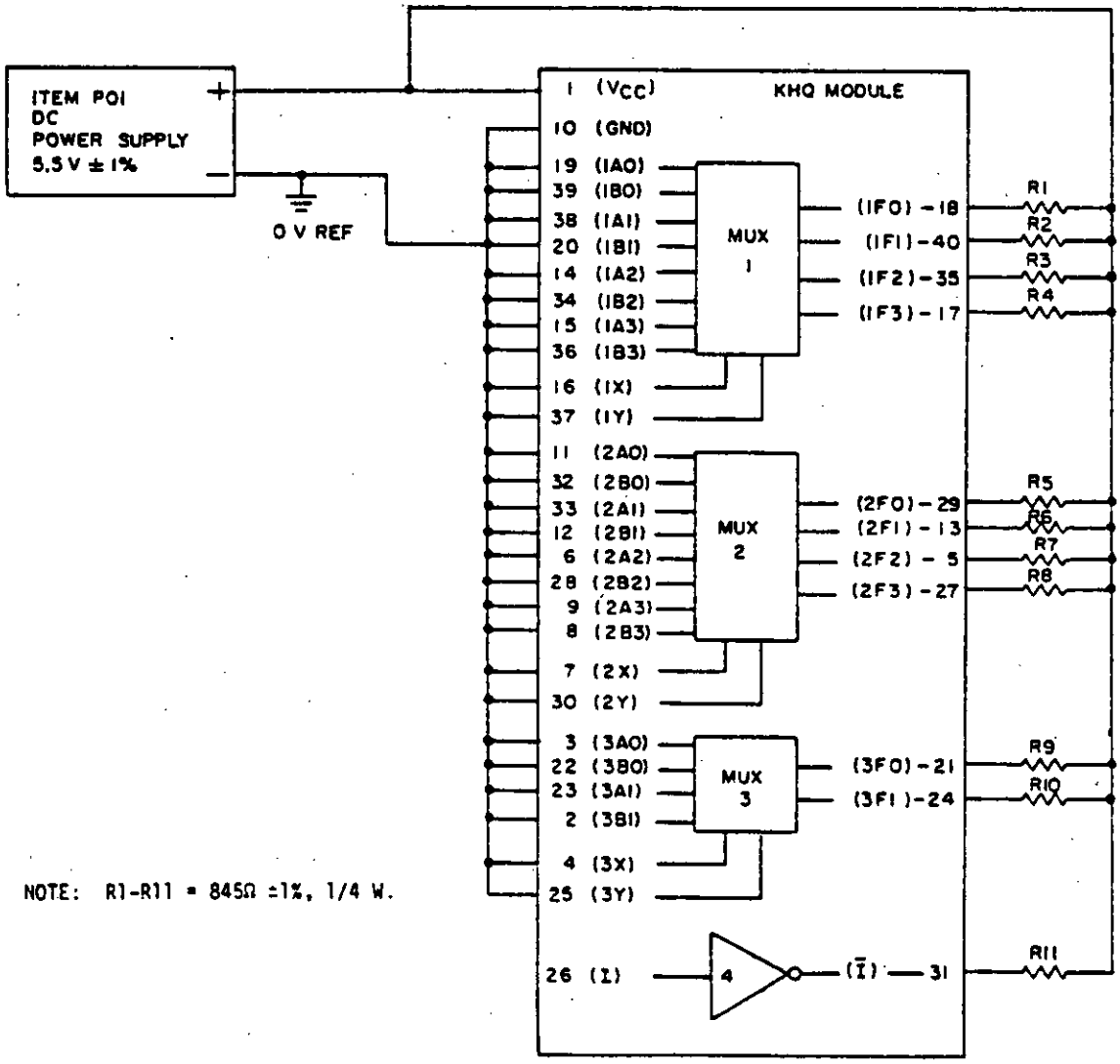


FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

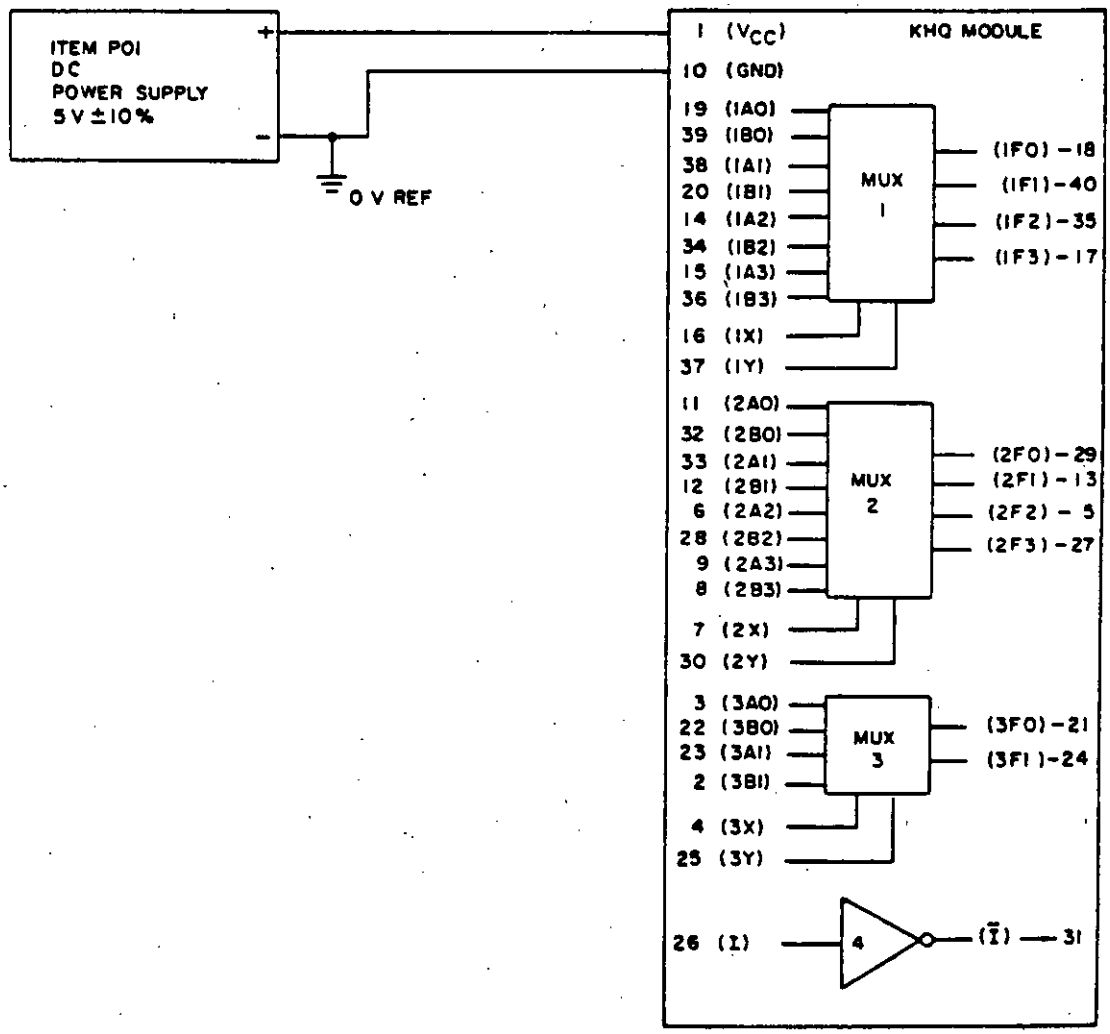


FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

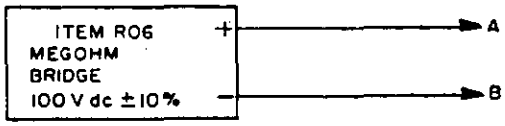
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1F0 (V_{OUT} (low loaded)	18	0.01	1.00	V	6
B2	1F1 "	40	"	"	"	"
B3	1F2 "	35	"	"	"	"
B4	1F3 "	17	"	"	"	"
B5	2F0 "	29	"	"	"	"
B6	2F1 "	13	"	"	"	"
B7	2F2 "	5	"	"	"	"
B8	2F3 "	27	"	"	"	"
B9	3F0 "	21	"	"	"	"
B10	3F1 "	24	"	"	"	"
B11	$\overline{T}$ "	31	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10 20T15:53Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

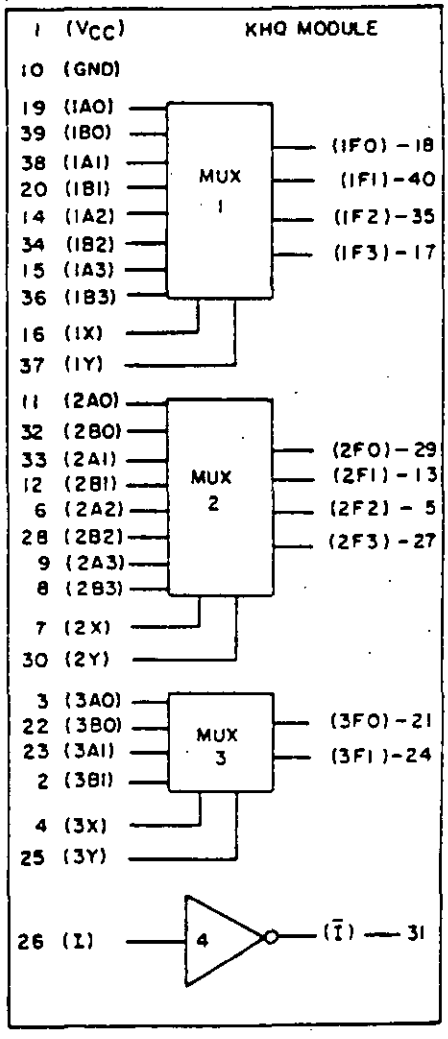
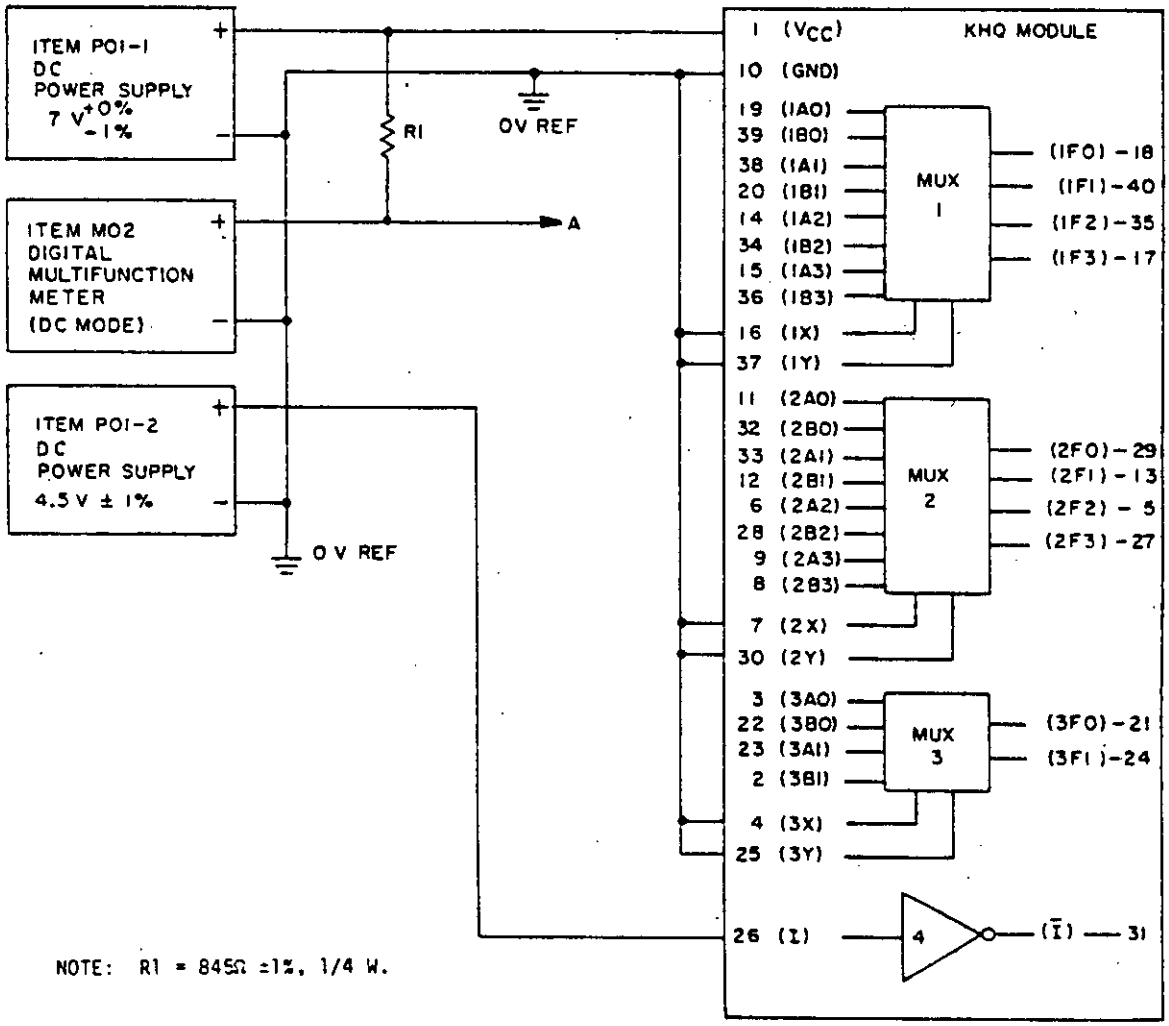


FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	1A0 Input breakdown current	19	39,38,20,14,34,15,36, 16,37,11,32,33,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	Open	0	100	µA	7
C2	1A1 "	38	19,39,20,14,34,15,36, 16,37,11,32,33,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C3	1A2 "	14	19,39,38,20,34,15,36, 16,37,11,32,33,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C4	1A3 "	15	19,39,38,20,14,34,36, 16,37,11,32,33,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C5	2A0 "	11	19,39,38,20,14,34,15, 36,16,37,32,33,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C6	2A1 "	33	19,39,38,20,14,34,15, 36,16,37,11,32,12,6, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C7	2A2 "	6	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 28,9,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"
C8	2A3 "	9	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 6,28,8,7,30,3,22,23, 2,4,25,26	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C9	3A0 Input breakdown current	3	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 6,28,9,8,7,30,22,23, 2,4,25,26	Open	0	100	µA	7
C10	3A1 "	23	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 6,28,9,8,7,30,3,22, 2,4,25,26	"	"	"	"	"
C11	3X "	4	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,25,26	"	"	"	"	"
C12	3Y "	25	19,39,38,20,14,34,15, 36,16,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,26	"	"	"	"	"
C13	I "	26	Open	"	"	"	"	"
C14	1B0 "	39	19,38,20,14,34,15, 36,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	16	"	200	"	"
C15	1B1 "	20	19,39,38,14,34,15, 36,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	"	"	"	"	"
C16	1B2 "	34	19,39,38,20,14,15, 36,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	"	"	"	"	"
C17	1B3 "	36	19,39,38,20,14,34, 15,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

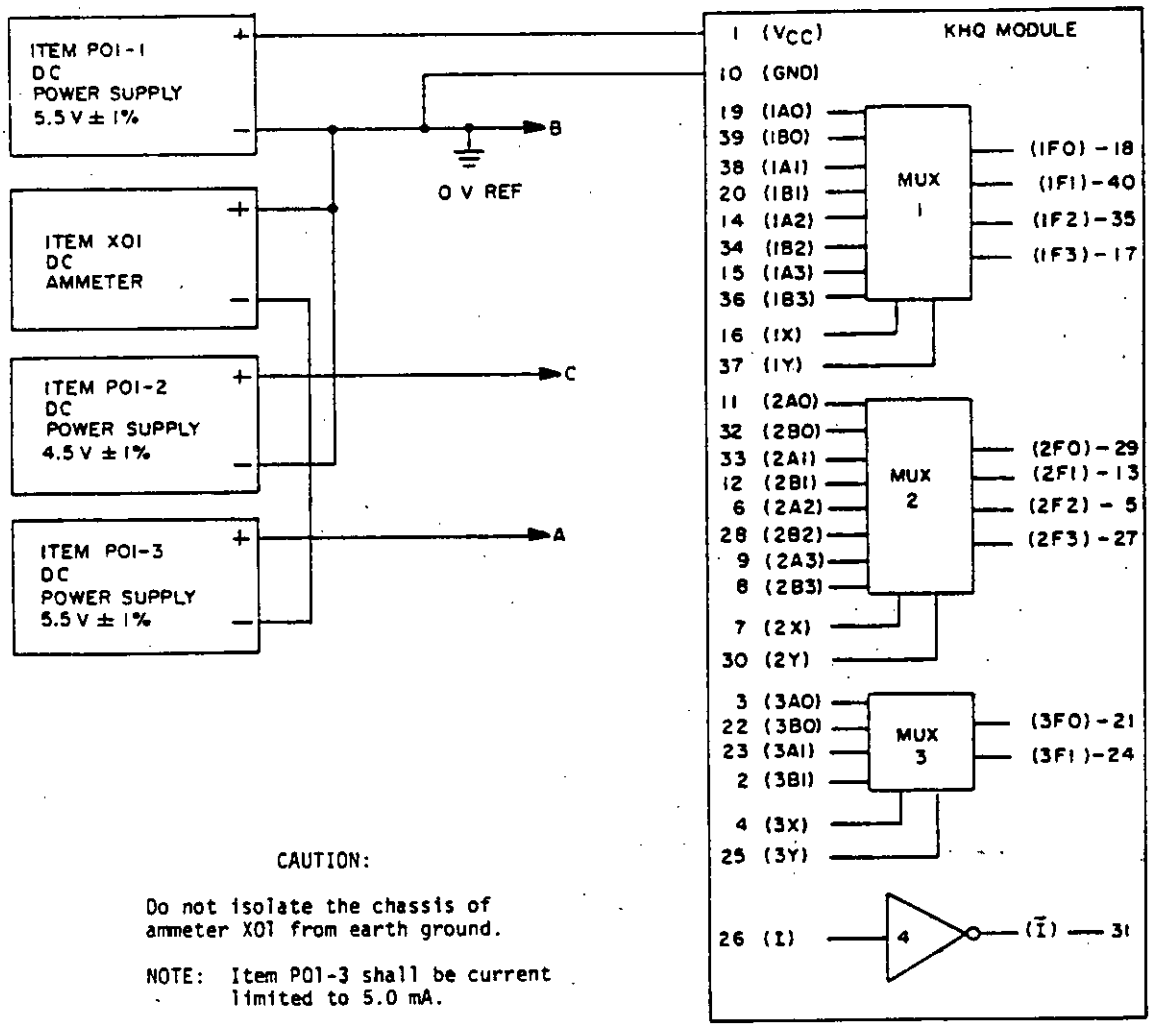
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C18	1X Input breakdown current	16	19,39,38,20,14,34, 15,36,37,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	Open	0	200	µA	7
C19	1Y "	37	19,39,38,20,14,34, 15,36,16,11,32,33,12, 6,28,9,8,7,30,3,22, 23,2,4,25,26	"	"	"	"	"
C20	2X "	7	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,28,9,8,30,3,22, 23,2,4,25,26	"	"	"	"	"
C21	2Y "	30	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,28,9,8,7,3,22, 23,2,4,25,26	"	"	"	"	"
C22	2B0 "	32	19,39,38,20,14,34, 15,36,16,37,11,33, 12,6,28,9,8,30,3,22, 23,2,4,25,26	7	"	"	"	"
C23	2B1 "	12	19,39,38,20,14,34, 15,36,16,37,11,32, 33,6,28,9,8,30,3,22, 23,2,4,25,26	"	"	"	"	"
C24	2B2 "	28	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,9,8,30,3,22, 23,2,4,25,26	"	"	"	"	"
C25	2B3 "	8	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,28,9,30,3,22, 23,2,4,25,26	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C26	3B0 Input breakdown current	22	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,28,9,8,7,30, 3,23,2,4,26	25	0	200	µA	7
C27	3B1 Input breakdown current	2	19,39,38,20,14,34, 15,36,16,37,11,32,33, 12,6,28,9,8,7,30, 3,22,23,4,26	25	0	200	µA	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Item PO1-3 shall be current limited to 5.0 mA.

FIGURE 7. Input breakdown current test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D1	1F0 Output short circuit current	Open	Open	19	18	38,14, 15,11, 33,6,9, 3,23	-6	-135	mA	8
D2	1F1 "	"	"	38	40	19,14, 15,11, 33,6,9, 3,23	"	"	"	"
D3	1F2 "	"	"	14	35	19,38, 15,11, 33,6,9, 3,23	"	"	"	"
D4	1F3 "	"	"	15	17	19,38, 14,11, 33,6,9, 3,23	"	"	"	"
D5	2F0 "	"	"	11	29	19,38, 14,15, 33,6,9, 3,23	"	"	"	"
D6	2F1 "	"	"	33	13	19,38, 14,15, 11,6,9, 3,23	"	"	"	"
D7	2F2 "	"	"	6	5	19,38, 14,15, 11,33, 9,3,23	"	"	"	"
D8	2F3 "	"	"	9	27	19,38, 14,15, 11,33, 6,3,23	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260420T15:53Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

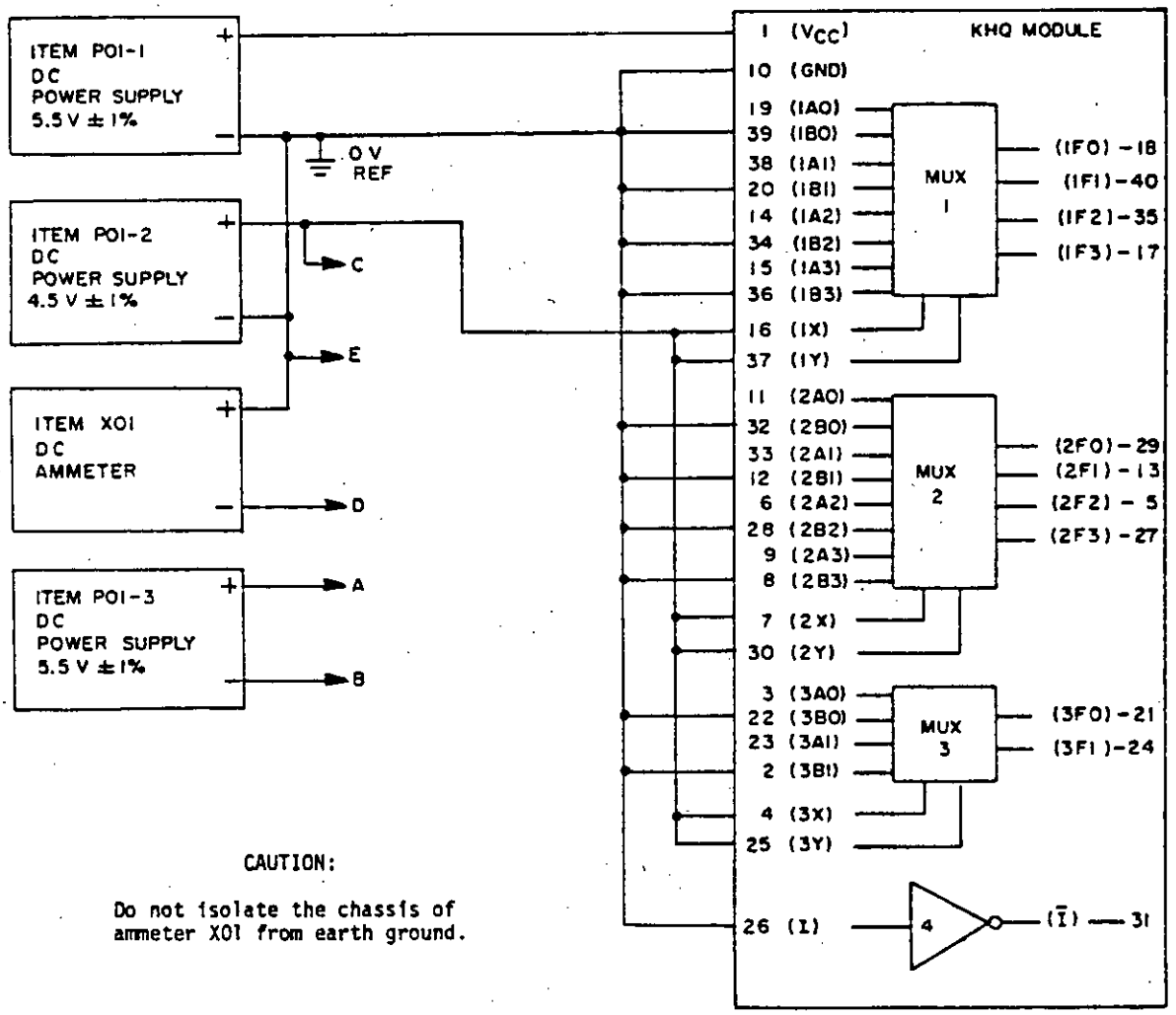
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
D9	3FO Output short circuit current	Open	Open	3	21	19,38, 14,15, 11,33, 6,9,23	-6	-135	mA	8
D10	3F1 "	"	"	23	24	19,38, 14,15, 11,33, 6,9,3	"	"	"	"
D11	$\overline{1}$ "	"	"	Open	31	19,38, 14,15, 11,33, 6,9,3, 23	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE VII. Output leakage current test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	IFO Output leakage current	18	0	19,38, 14,15, 11,33, 6,9,3, 23	B	Open	0	250	µA	8
E2	IF1 "	40	"	"	"	"	"	"	"	"
E3	IF2 "	35	"	"	"	"	"	"	"	"
E4	IF3 "	17	"	"	"	"	"	"	"	"
E5	2F0 "	29	"	"	"	"	"	"	"	"
E6	2F1 "	13	"	"	"	"	"	"	"	"
E7	2F2 "	5	"	"	"	"	"	"	"	"
E8	2F3 "	27	"	"	"	"	"	"	"	"
E9	3F0 "	21	"	"	"	"	"	"	"	"
E10	3F1 "	24	"	"	"	"	"	"	"	"
E11	T "	31	"	"	"	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 1 second.
2. Item PO1-3 shall be current limited to 5.0 mA.

FIGURE 8. Output short circuit current and output leakage current test circuit for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1F0 Output voltage, high level	16,19, 37,26	39,20,38, 34,14,36, 15,7,30, 4,25	18	2.5	4.5	V	9
F2	1F1 "	16,37, 38,26	39,19,20, 34,14,36, 15,7,30, 4,25	40	"	"	"	"
F3	1F2 "	16,37, 14,26	39,19,20, 38,34,36, 15,7,30, 4,25	35	"	"	"	"
F4	1F3 "	16,37, 15,26	39,19,20, 38,34,14, 36,7,30, 4,25	17	"	"	"	"
F5	2F0 "	7,30, 11,26	32,12,33, 28,6,8,9, 16,37,4, 25	29	"	"	"	"
F6	2F1 "	7,30, 33,26	32,11,12, 28,6,8,9, 16,37,4, 25	13	"	"	"	"
F7	2F2 "	7,30, 6,26	32,11,12, 33,28,8, 9,16,37, 4,25	5	"	"	"	"
F8	2F3 "	7,30, 9,26	32,11,12, 33,28,6, 8,16,37, 4,25	27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE VIII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F9	3F0 Output voltage, high level	4,25, 3,26	22,2,23, 16,37,7, 30	21	2.5	4.5	V	9
F10	3F1 "	4,25, 23,26	22,3,2, 16,37,7, 30	24	"	"	"	"
F11	T "	Open	26	31	"	"	"	"
F12	1F0 Output voltage, low level	39,19,20 38,34,14, 36,15	16,37	18	0	500	mV	"
F13	1F1 "	"	"	40	"	"	"	"
F14	1F2 "	"	"	35	"	"	"	"
F15	1F3 "	"	"	17	"	"	"	"
F16	2F0 "	32,11,12, 33,28,6, 8,9	7,30	29	"	"	"	"
F17	2F1 "	"	"	13	"	"	"	"
F18	2F2 "	"	"	5	"	"	"	"
F19	2F3 "	"	"	27	"	"	"	"
F20	3F0 "	22,3,2,23	4,25	21	"	"	"	"
F21	3F1 "	22,3,2,23	4,25	24	"	"	"	"
F22	T "	26	Open	31	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE IX. Functional test.

Test condition	Parameter	Connect			Output logic status 1/				Test figure
		A to pin	B to pin	C to pin	IF0	IF1	IF2	IF3	
G1	Functional, MUX 1	38,15,16,37	19,39,20,14,34,36	2/ 3/	L	H	L	H	9
G2	"	19,39,20,14,34,36, 16,37	38,15	"	H	L	H	L	"
G3	"	19,39,38,14,34,15,37	20,36,16	"	H	L	H	L	"
G4	"	19,38,20,14,15,36,16	39,34,37	"	H	L	H	L	"
G5	"	19,39,38,20,14,34, 15,36	16,37	"	L	L	L	L	"
G6	"	19,14,16,37	39,38,20,34,15,36	"	H	L	H	L	"
G7	"	39,38,20,34,15,36, 16,37	19,14	"	L	H	L	H	"
G8	"	19,38,20,14,15,36,37	39,34,16	"	L	H	L	H	"
G9	"	19,39,38,14,34,15,16	20,36,37	"	L	H	L	H	"
G10	"	Open	19,39,38,20,14,34, 15,36,16,37	"	L	L	L	L	"
G11	Functional, MUX 2	33,9,7,30	11,32,12,6,28,8	"	L	H	L	H	"
G12	"	11,32,12,6,28,8,7,30	33,9	"	H	L	H	L	"
G13	"	11,32,33,6,28,9,30	12,8,7	"	H	L	H	L	"
G14	"	11,33,12,6,9,8,7	32,28,30	"	H	L	H	L	"
G15	"	11,32,33,12,6,28,9,8	7,30	"	L	L	L	L	"
G16	"	11,6,7,30	32,33,12,28,9,8	"	H	L	H	L	"
G17	"	32,33,12,28,9,8,7,30	11,6	"	L	H	L	H	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE IX. Functional test - Continued.

Test condition	Parameter	Connect			Output logic status 1/				Test figure
		A to pin	B to pin	C to pin	IF0	IF1	IF2	IF3	
G18	Functional, MUX 2	11,33,12,6,9,8,30	32,28,7	2/ 3/	L	H	L	H	9
G19	"	11,32,33,6,28,9,7	12,8,30	"	L	H	L	H	"
G20	"	Open	11,32,33,12,6,28, 9,8,7,30	"	L	L	L	L	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE IX. Functional test - Continued.

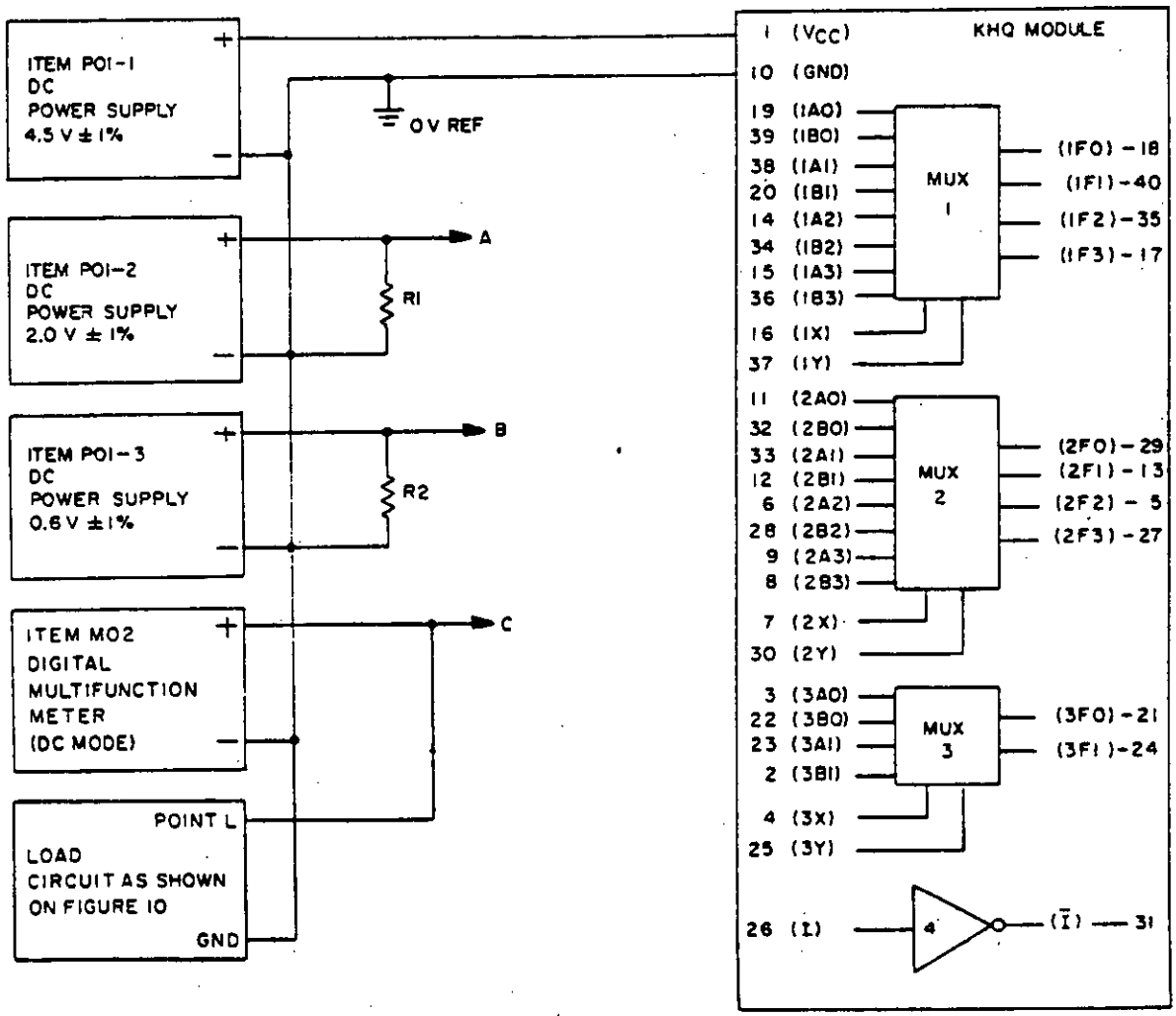
Test condition	Parameter	Connect			Output logic status 1/			Test figure
		A to pin	B to pin	C to pin	3F0	3F1	$\overline{T}$	
G21	Functional, MUX 3	23,4,25	3,22,2	2/ 3/	L	H		9
G22	"	3,22,2,4,25	23	"	H	L		"
G23	"	3,22,23,25	2,4	"	H	L		"
G24	"	3,23,2,4	22,25	"	H	L		"
G25	"	3,22,23,2	4,25	"	L	L		"
G26	"	3,4,25	22,23,2	"	H	L		"
G27	"	22,23,2,4,25	3	"	L	H		"
G28	"	3,23,2,25	22,4	"	L	H		"
G29	"	3,22,23,4	2,25	"	L	H		"
G30	"	Open	3,22,23,2,4,25	"	L	L		"
G31	Functional, INVERTER	26	Open	"			L	"
G32	Functional, INVERTER	Open	26	"			H	"

1/ Output logic states:
L = Low level = 0.0 to 0.5 V
H = High level = 2.5 to 4.5 V

2/ For test conditions G1-G32 verification of the output logic states in footnote 1 is obtained by connecting item M02, point C, to each output in succession after input conditions are accomplished.

3/ 1F0-1F3, 2F0-2F3, 3F0, 3F1, and $\overline{T}$ output logic states are measured at output pins 18, 40, 35, 17, 29, 13, 5, 27, 21, 24, and 31, respectively.

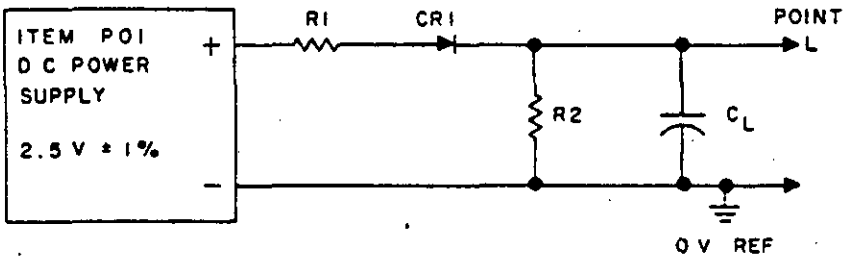
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 27Ω ±5%, 1/4 W.

* FIGURE 9. Output voltage (high-level and low-level) and functional test circuit for tables VIII and IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 6 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance) except for output crosstalk test which will use 50 ±5 pF.
5. Capacitor C_L is not required for dc electrical test conditions.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 10. KHQ logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE X. Input clamp diode voltage test.

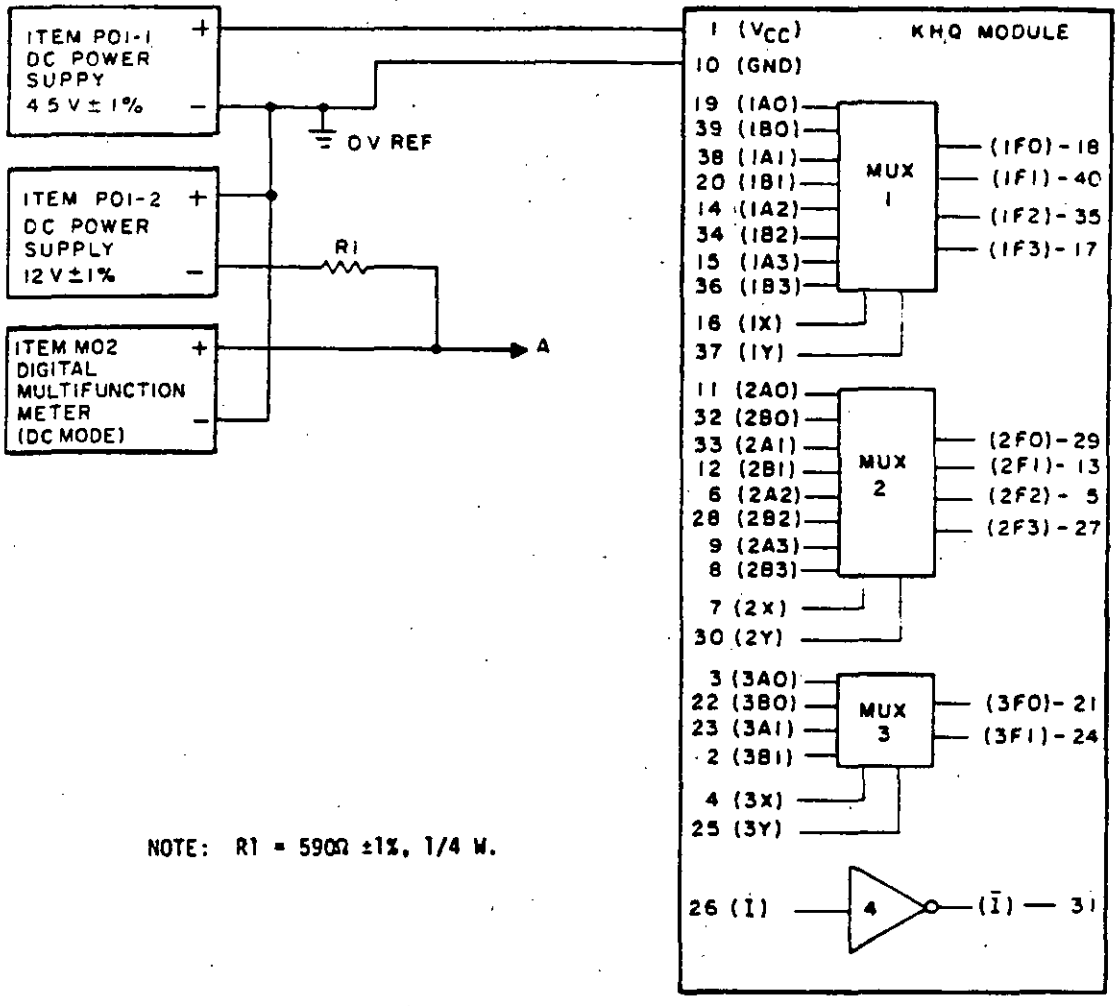
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
H1	1A0 Input clamp diode voltage	19	-0.4	-1.5	V	11
H2	1B0 "	39	"	"	"	"
H3	1A1 "	38	"	"	"	"
H4	1B1 "	20	"	"	"	"
H5	1A2 "	14	"	"	"	"
H6	1B2 "	34	"	"	"	"
H7	1A3 "	15	"	"	"	"
H8	1B3 "	36	"	"	"	"
H9	1X "	16	"	"	"	"
H10	1Y "	37	"	"	"	"
H11	2A0 "	11	"	"	"	"
H12	2B0 "	32	"	"	"	"
H13	2A1 "	33	"	"	"	"
H14	2B1 "	12	"	"	"	"
H15	2A2 "	6	"	"	"	"
H16	2B2 "	28	"	"	"	"
H17	2A3 "	9	"	"	"	"
H18	2B3 "	8	"	"	"	"
H19	2X "	7	"	"	"	"
H20	2Y "	30	"	"	"	"
H21	3A0 "	3	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE X. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
H22	3B0 Input clamp diode voltage	22	-0.4	-1.5	V	11
H23	3A1 "	23	"	"	"	"
H24	3B1 "	2	"	"	"	"
H25	3X "	4	"	"	"	"
H26	3Y "	25	"	"	"	"
H27	$\overline{T}$ "	26	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 11. Input clamp diode voltage test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A0 Input current, high level	2,3,4,6,7,8,9,11,12,14, 15,16,20,22,23,25,26, 28,30,32,33,34,36,37, 38,39	19	0	20	µA	12
J2	1A1 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,39	38	"	"	"	"
J3	1A2 "	2,3,4,6,7,8,9,11,12,15, 16,19,20,22,23,25,26, 28,30,32,33,34,36,37, 38,39	14	"	"	"	"
J4	1A3 "	2,3,4,6,7,8,9,11,12,14, 16,19,20,22,23,25,26, 28,30,32,33,34,36,37, 38,39	15	"	"	"	"
J5	2A0 "	2,3,4,6,7,8,9,12,14,15, 16,19,20,22,23,25,26, 28,30,32,33,34,36,37, 38,39	11	"	"	"	"
J6	2A1 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,34,36,37, 38,39	33	"	"	"	"
J7	2A2 "	2,3,4,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	6	"	"	"	"
J8	2A3 "	2,3,4,6,7,8,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	9	"	"	"	"
J9	3A0 "	2,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	3	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J10	3A1 Input current, high level	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,25,26, 28,30,32,33,34,36,37, 38,39	23	0	20	µA	12
J11	3X "	2,3,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	4	"	"	"	"
J12	3Y "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23, 26,28,30,32,33,34,36, 37,38,39	25	"	"	"	"
J13	I "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23, 25,28,30,32,33,34,36, 37,38,39	26	"	"	"	"
J14	IB0 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,	39	"	40	"	"
J15	IB1 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,22,23,25, 26,28,30,32,33,34,36, 37,38,39	20	"	"	"	"
J16	IB2 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,36, 37,38,39	34	"	"	"	"
J17	IB3 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34, 37,38,39	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

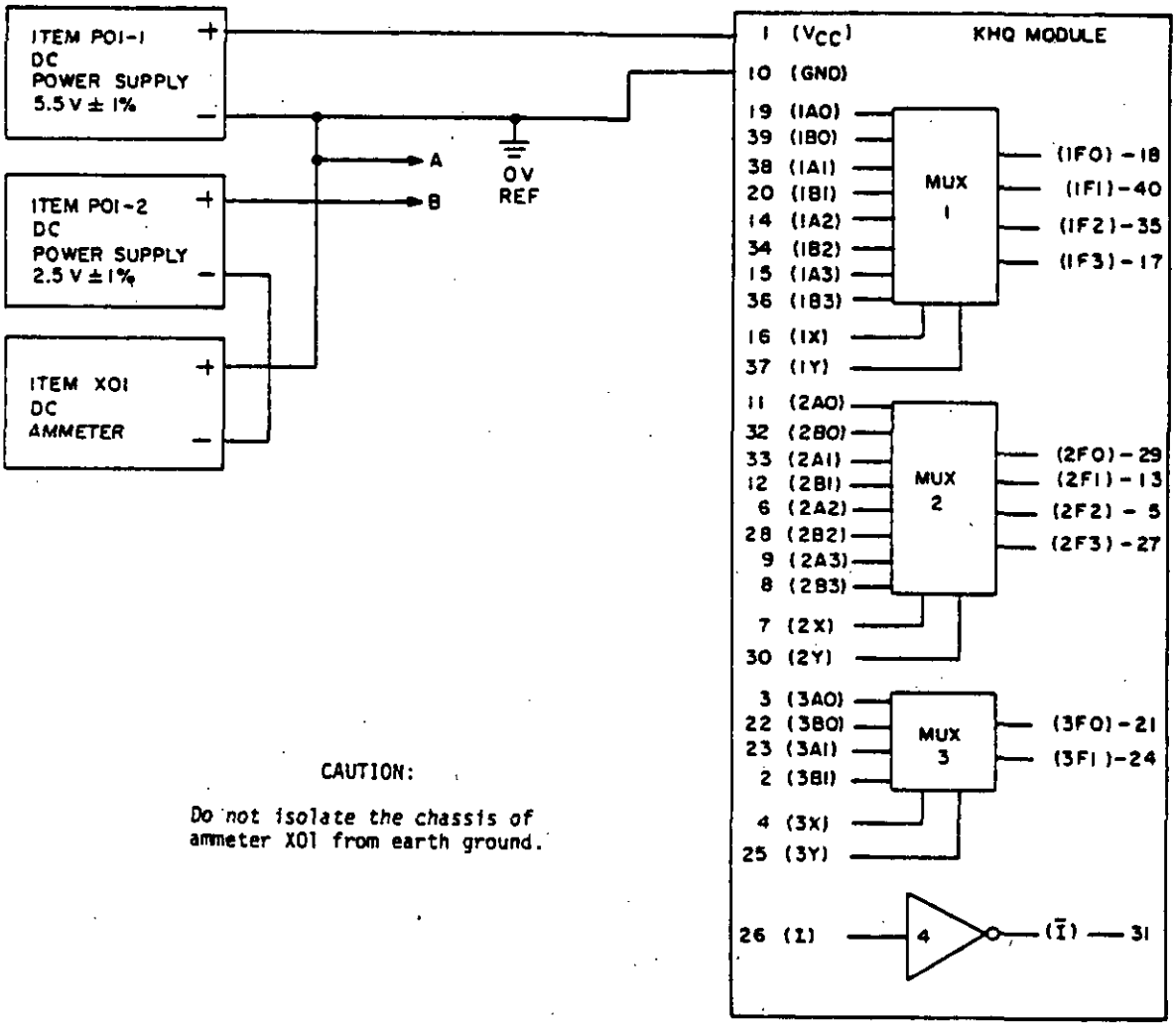
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J18	1X Input current, high level	2,3,4,6,7,8,9,11,12,14, 15,19,20,22,23,25,26, 28,30,32,33,34,36,37, 38,39	16	0	40	µA	12
J19	1Y "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34, 36,38,39	37	"	"	"	"
J20	2B0 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,33,34,36, 37,38,39	32	"	"	"	"
J21	2B1 "	2,3,4,6,7,8,9,11,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	12	"	"	"	"
J22	2B2 "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,30,32,33,34,36, 37,38,39	28	"	"	"	"
J23	2B3 "	2,3,4,6,7,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	8	"	"	"	"
J24	2X "	2,3,4,6,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	7	"	"	"	"
J25	2Y "	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,32,33,34,36, 37,38,39	30	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J26	3B0 Input current, high level	2,3,4,6,7,8,9,11,12,14, 15,16,19,20,23,25,26 28,30,32,33,34,36,37, 38,39	22	0	40	µA	12
J27	3B1 Input current, high level	3,4,6,7,8,9,11,12,14, 15,16,19,20,22,23,25, 26,28,30,32,33,34,36, 37,38,39	2	0	40	µA	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 12. Input current (high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1A0 Input current, low level	16,37	Open	19	-0.10	-0.44	mA	13
K2	1A1 "	"	"	38	"	"	"	"
K3	1A2 "	"	"	14	"	"	"	"
K4	1A3 "	"	"	15	"	"	"	"
K5	2A0 "	7,30	"	11	"	"	"	"
K6	2A1 "	"	"	33	"	"	"	"
K7	2A2 "	"	"	6	"	"	"	"
K8	2A3 "	"	"	9	"	"	"	"
K9	3A0 "	4,25	"	3	"	"	"	"
K10	3A1 "	4,25	"	23	"	"	"	"
K11	3X "	Open	"	4	"	"	"	"
K12	3Y "	"	"	25	"	"	"	"
K13	1 "	"	"	26	"	"	"	"
K14	1B0 "	37	16	39	-0.20	-0.88	"	"
K15	1B1 "	"	"	20	"	"	"	"
K16	1B2 "	"	"	34	"	"	"	"
K17	1B3 "	"	"	36	"	"	"	"
K18	2B0 "	30	7	32	"	"	"	"
K19	2B1 "	"	"	12	"	"	"	"
K20	2B2 "	"	"	28	"	"	"	"
K21	2B3 "	"	"	8	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K22	3B0 Input current, low level	25	4	22	-0.20	-0.88	mA	13
K23	3B1 "	25	4	2	"	"	"	"
K24	2X "	Open	Open	7	"	"	"	"
K25	2Y "	"	"	30	"	"	"	"
K26	1X "	"	"	16	"	"	"	"
K27	1Y "	"	"	37	"	"	"	"

TABLE XIII. Power supply current (low-level outputs) test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current, low level outputs	10	64	mA	14

Source: https://assist.dla.mil -- Downloaded: 20261020T15:53Z
Check the source to verify that this is the current version before use.

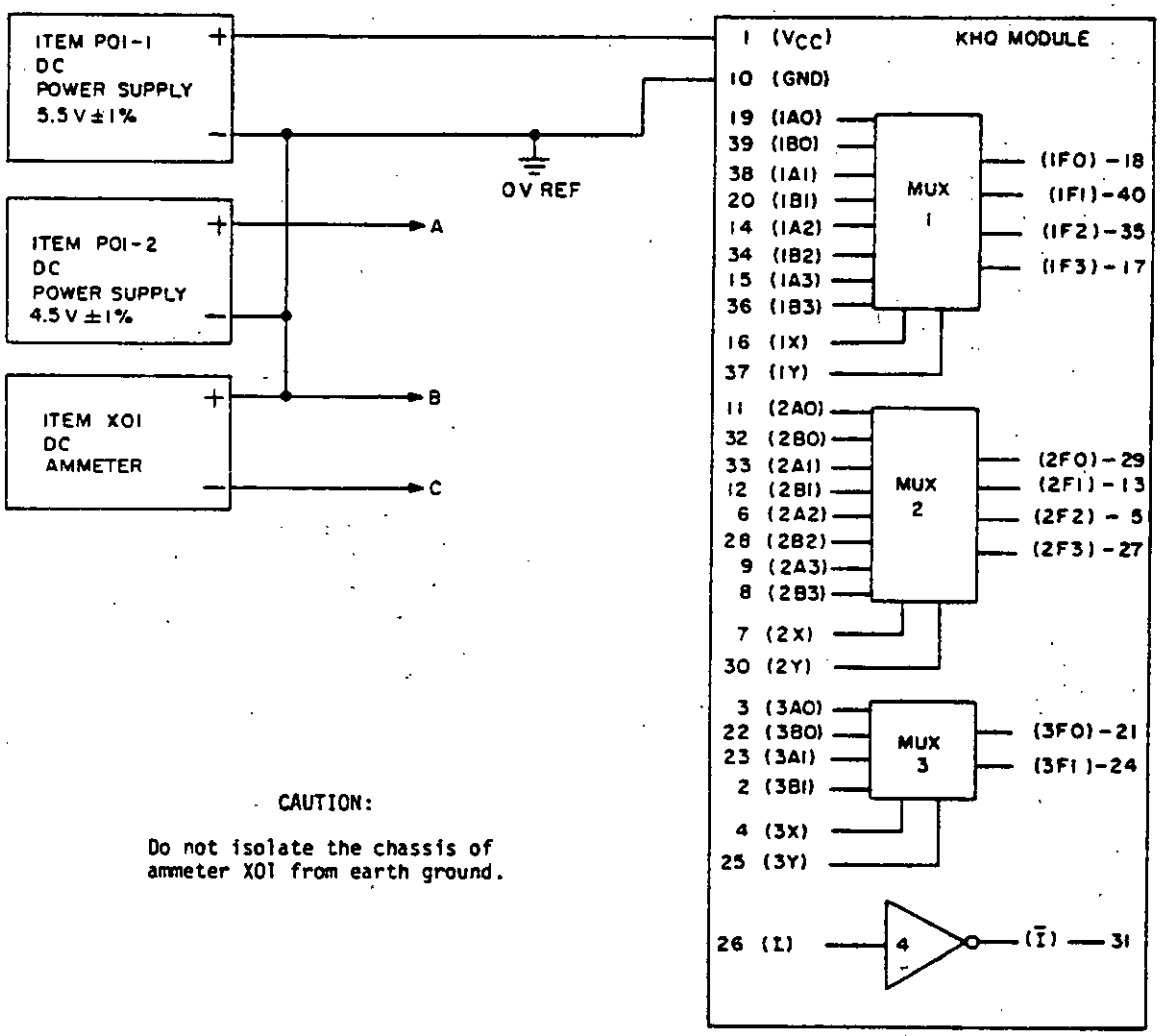


FIGURE 13. Input current (low-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

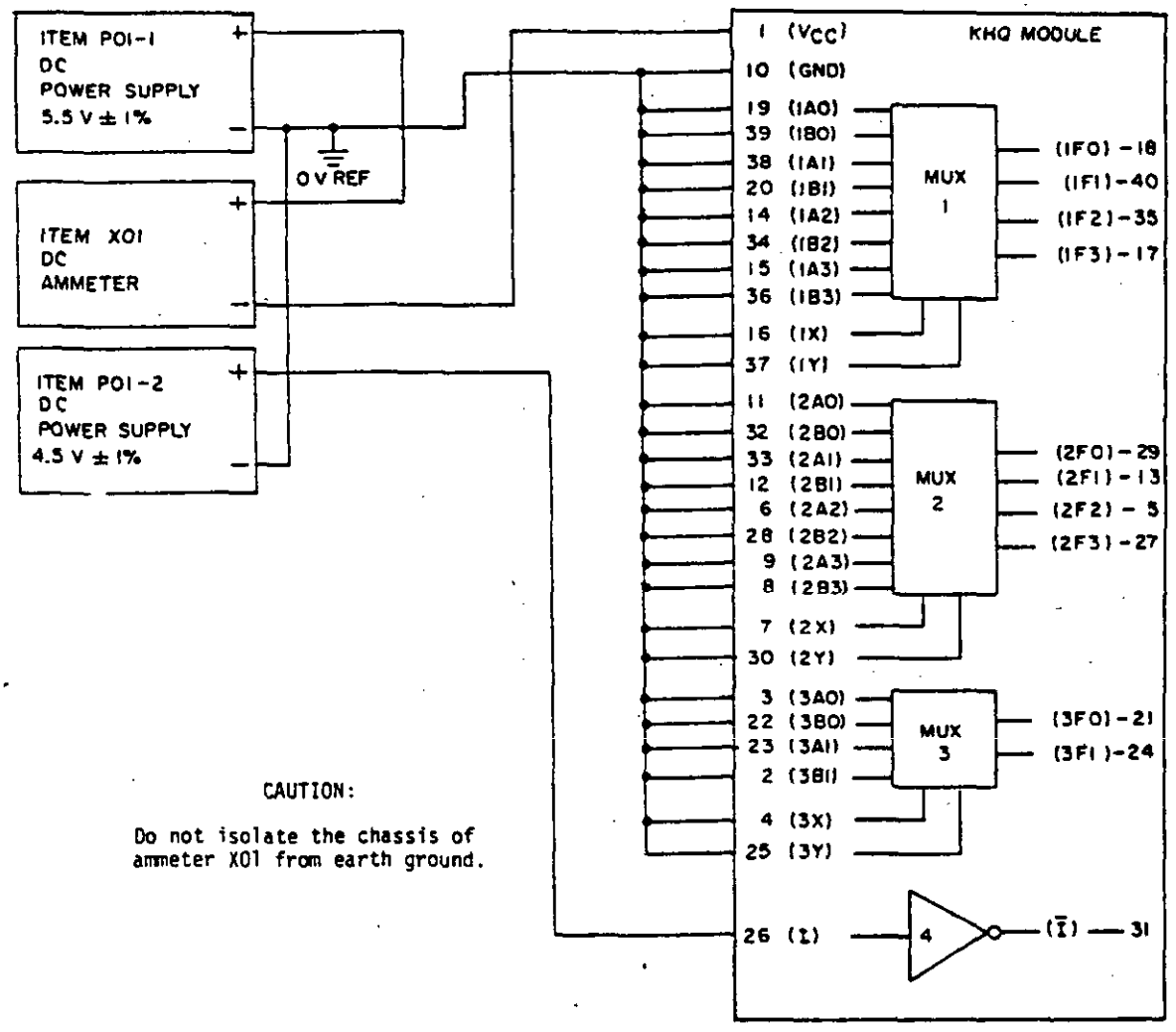


FIGURE 14. Power supply current (low-level outputs) test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XIV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1F0 Transition time (t_{TLH}), C_L = 50 pF	16,37	Open	19	18	10	50	ns	15,16
M2	1F1 "	"	"	38	40	"	"	"	"
M3	1F2 "	"	"	14	35	"	"	"	"
M4	1F3 "	"	"	15	17	"	"	"	"
M5	2F0 "	7,30	"	11	29	"	"	"	"
M6	2F1 "	"	"	33	13	"	"	"	"
M7	2F2 "	"	"	6	5	"	"	"	"
M8	2F3 "	"	"	9	27	"	"	"	"
M9	3F0 "	4,25	"	3	21	"	"	"	"
M10	3F1 "	4,25	"	23	24	"	"	"	"
M11	$\overline{I}$ "	Open	"	26	31	"	"	"	15,17
M12	1F0 Transition time (t_{TLH}), C_L = 150 pF	16,37	"	19	18	20	80	"	15,16
M13	1F1 "	"	"	38	40	"	"	"	"
M14	1F2 "	"	"	14	35	"	"	"	"
M15	1F3 "	"	"	15	17	"	"	"	"
M16	2F0 "	7,30	"	11	29	"	"	"	"
M17	2F1 "	"	"	33	13	"	"	"	"
M18	2F2 "	"	"	6	5	"	"	"	"
M19	2F3 "	"	"	9	27	"	"	"	"
M20	3F0 "	4,25	"	3	21	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M21	3F1 Transition time (t_{TLH}), C_L = 150 pF	4,25	Open	23	24	20	80	ns	15,16
M22	T Transition time (t_{TLH}), C_L =150 pF	Open	"	26	31	20	80	"	15,17
N1	1F0 Transition time (t_{THL}), C_L = 50 pF	16,37	"	19	18	3	18	"	15,16
N2	1F1 "	"	"	38	40	"	"	"	"
N3	1F2 "	"	"	14	35	"	"	"	"
N4	1F3 "	"	"	15	17	"	"	"	"
N5	2F0 "	7,30	"	11	29	"	"	"	"
N6	2F1 "	"	"	33	13	"	"	"	"
N7	2F2 "	"	"	6	5	"	"	"	"
N8	2F3 "	"	"	9	27	"	"	"	"
N9	3F0 "	4,25	"	3	21	"	"	"	"
N10	3F1 "	4,25	"	23	24	"	"	"	"
N11	T "	Open	"	26	31	"	"	"	15,17
N12	1F0 Transition time (t_{THL}), C_L = 150 pF	16,37	"	19	18	6	40	"	15,16
N13	1F1 "	"	"	38	40	"	"	"	"
N14	1F2 "	"	"	14	35	"	"	"	"
N15	1F3 "	"	"	15	17	"	"	"	"
N16	2F0 "	7,30	"	11	29	"	"	"	"
N17	2F1 "	7,30	"	33	13	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260420T15:53Z
Check the source to verify that this is the current version before use.

TABLE XIV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N18	2F2 Transition time (t_{THL}), C_L = 150 pF	7,30	Open	6	5	6	40	ns	15,16
N19	2F3 "	7,30	"	9	27	"	"	"	"
N20	3F0 "	4,25	"	3	21	"	"	"	"
N21	3F1 "	4,25	"	23	24	"	"	"	"
N22	T "	Open	"	26	31	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1A0 Propagation delay time (t_{PHL}), A0-A3 to output C_L = 50 pF	16,37	Open	19	18	5	31	ns	15,16
P2	1A1 "	"	"	38	40	"	"	"	"
P3	1A2 "	"	"	14	35	"	"	"	"
P4	1A3 "	"	"	15	17	"	"	"	"
P5	2A0 "	7,30	"	11	29	"	"	"	"
P6	2A1 "	"	"	33	13	"	"	"	"
P7	2A2 "	"	"	6	5	"	"	"	"
P8	2A3 "	"	"	9	27	"	"	"	"
P9	3A0 "	4,25	"	3	21	"	"	"	"
P10	3A1 "	4,25	"	23	24	"	"	"	"
P11	1B0 Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 50 pF	37	16	39	18	"	"	"	"
P12	1B1 "	"	"	20	40	"	"	"	"
P13	1B2 "	"	"	34	35	"	"	"	"
P14	1B3 "	"	"	36	17	"	"	"	"
P15	2B0 "	30	7	32	29	"	"	"	"
P16♠	2B1 "	"	"	12	13	"	"	"	"
P17	2B2 "	"	"	28	5	"	"	"	"
P18	2B3 "	"	"	8	27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10/20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P19	380	Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 50 pF	25	4	22	21	5	31	ns	15,16
P20	381	Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 50 pF	25	4	2	24	"	31	"	15,16
P21	180	Propagation delay time (t_{PHL}), B0-B3 to output (inverting), C_L = 50 pF	16	37	39	18	"	46	"	15,17
P22	181	"	"	"	20	40	"	"	"	"
P23	182	"	"	"	34	35	"	"	"	"
P24	183	"	"	"	36	17	"	"	"	"
P25	280	"	7	30	32	29	"	"	"	"
P26	281	"	"	"	12	13	"	"	"	"
P27	282	"	"	"	28	5	"	"	"	"
P28	283	"	"	"	8	27	"	"	"	"
P29	380	"	4	25	22	21	"	"	"	"
P30	381	"	4	25	2	24	"	46	"	15,17
P31	1X to 1F0	Propagation delay time (t_{PHL}), X and Y address to output, C_L = 50 pF	37,19	39	16	18	"	43	"	15,16
P32	1X to 1F0	"	37,39	19	"	18	"	"	"	15,17
P33	1X to 1F1	"	37,38	20	"	40	"	"	"	15,16
P34	1X to 1F1	"	37,20	38	"	40	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P35	1X to 1F2 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 50 pF	37,14	34	16	35	5	43	ns	15,16
P36	1X to 1F2 "	37,34	14	"	35	"	"	"	15,17
P37	1X to 1F3 "	37,15	36	"	17	"	"	"	15,16
P38	1X to 1F3 "	37,36	15	"	17	"	"	"	15,17
P39	1Y to 1F0 "	39,16,19	Open	37	18	"	"	"	15,16
P40	1Y to 1F0 "	16	19,39	"	18	"	"	"	15,17
P41	1Y to 1F1 "	20,16,38	Open	"	40	"	"	"	15,16
P42	1Y to 1F1 "	16	20,38	"	40	"	"	"	15,17
P43	1Y to 1F2 "	34,16,14	Open	"	35	"	"	"	15,16
P44	1Y to 1F2 "	16	14,34	"	35	"	"	"	15,17
P45	1Y to 1F3 "	36,16,15	Open	"	17	"	"	"	15,16
P46	1Y to 1F3 "	16	15,36	"	17	"	"	"	15,17
P47	2X to 2F0 "	30,11	32	7	29	"	"	"	15,16
P48	2X to 2F0 "	30,32	11	"	29	"	"	"	15,17
P49	2X to 2F1 "	30,33	12	"	13	"	"	"	15,16
P50	2X to 2F1 "	30,12	33	"	13	"	"	"	15,17
P51	2X to 2F2 "	30,6	28	"	5	"	"	"	15,16
P52	2X to 2F2 "	30,28	6	"	5	"	"	"	15,17
P53	2X to 2F3 "	30,9	8	"	27	"	"	"	15,16
P54	2X to 2F3 "	30,8	9	"	27	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 2026/10-20 15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P55	2Y to 2F0 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 50 pF	32,7,11	Open	30	29	5	43	ns	15,16
P56	2Y to 2F0 "	7	11,32	"	29	"	"	"	15,17
P57	2Y to 2F1 "	12,7,33	Open	"	13	"	"	"	15,16
P58	2Y to 2F1 "	7	12,33	"	13	"	"	"	15,17
P59	2Y to 2F2 "	28,7,6	Open	"	5	"	"	"	15,16
P60	2Y to 2F2 "	7	6,28	"	5	"	"	"	15,17
P61	2Y to 2F3 "	8,7,9	Open	"	27	"	"	"	15,16
P62	2Y to 2F3 "	7	8,9	"	27	"	"	"	15,17
P63	3X to 3F0 "	25,3	22	4	21	"	"	"	15,16
P64	3X to 3F0 "	25,22	3	"	21	"	"	"	15,17
P65	3X to 3F1 "	25,23	2	"	24	"	"	"	15,16
P66	3X to 3F1 "	25,2	23	"	24	"	"	"	15,17
P67	3Y to 3F0 "	22,4,3	Open	25	21	"	"	"	15,16
P68	3Y to 3F0 "	4	3,22	"	21	"	"	"	15,17
P69	3Y to 3F1 "	2,4,23	Open	"	24	"	"	"	15,16
P70	3Y to 3F1 "	4	2,23	"	24	"	"	"	15,17
P71	I to $\overline{T}$ Propagation delay time (t_{PHL}), INVERTER, C_L = 50 pF	Open	Open	26	31	3	19	"	15,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P72	1A0	Propagation delay time (t_{PHL}), A0-A3 to output, C_L = 150 pF	16,37	Open	19	18	5	41	ns	15,16
P73	1A1	"	"	"	38	40	"	"	"	"
P74	1A2	"	"	"	14	35	"	"	"	"
P75	1A3	"	"	"	15	17	"	"	"	"
P76	2A0	"	7,30	"	11	29	"	"	"	"
P77	2A1	"	"	"	33	13	"	"	"	"
P78	2A2	"	"	"	6	5	"	"	"	"
P79	2A3	"	"	"	9	27	"	"	"	"
P80	3A0	"	4,25	"	3	21	"	"	"	"
P81	3A1	"	4,25	"	23	24	"	"	"	"
P82	1B0	Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 150 pF	37	16	39	18	"	"	"	"
P83	1B1	"	"	"	20	40	"	"	"	"
P84	1B2	"	"	"	34	35	"	"	"	"
P85	1B3	"	"	"	36	17	"	"	"	"
P86	2B0	"	30	7	32	29	"	"	"	"
P87	2B1	"	"	"	12	13	"	"	"	"
P88	2B2	"	"	"	28	5	"	"	"	"
P89	2B3	"	"	"	8	27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10/20T16:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P90	1B0 Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 150 pF	25	4	22	21	5	41	ns	15,16
P91	3B1 Propagation delay time (t_{PHL}), B0-B3 to output (noninverting), C_L = 150 pF	25	4	2	24	"	41	"	15,16
P92	1B0 Propagation delay time (t_{PHL}), B0-B3 to output (inverting), C_L = 150 pF	16	37	39	18	"	56	"	15,17
P93	1B1 "	"	"	20	40	"	"	"	"
P94	1B2 "	"	"	34	35	"	"	"	"
P95	1B3 "	"	"	36	17	"	"	"	"
P96	2B0 "	7	30	32	20	"	"	"	"
P97	2B1 "	"	"	12	13	"	"	"	"
P98	2B2 "	"	"	28	5	"	"	"	"
P99	2B3 "	"	"	8	27	"	"	"	"
P100	3B0 "	4	25	22	21	"	"	"	"
P101	3B1 "	4	25	2	24	"	"	"	"
P102	1X to 1F0 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	37,19	39	16	18	10	53	"	15,16
P103	1X to 1F0 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	37,39	19	16	18	10	53	"	15,17

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P104	1X to 1F1 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	37,38	20	16	40	10	53	ns	15,16
P105	1X to 1F1 "	37,20	38	"	40	"	"	"	15,17
P106	1X to 1F2 "	37,14	34	"	35	"	"	"	15,16
P107	1X to 1F2 "	37,34	14	"	35	"	"	"	15,17
P108	1X to 1F3 "	37,15	36	"	17	"	"	"	15,16
P109	1X to 1F3 "	37,36	15	"	17	"	"	"	15,17
P110	1Y to 1F0 "	39,16,19	Open	37	18	"	"	"	15,16
P111	1Y to 1F0 "	16	19,39	"	18	"	"	"	15,17
P112	1Y to 1F1 "	20,16,38	Open	"	40	"	"	"	15,16
P113	1Y to 1F1 "	16	20,38	"	40	"	"	"	15,17
P114	1Y to 1F2 "	34,16,14	Open	"	35	"	"	"	15,16
P115	1Y to 1F2 "	16	14,34	"	35	"	"	"	15,17
P116	1Y to 1F3 "	36,16,15	Open	"	17	"	"	"	15,16
P117	1Y to 1F3 "	16	15,36	"	17	"	"	"	15,17
P118	2X to 2F0 "	30,11	32	7	29	"	"	"	15,16
P119	2X to 2F0 "	30,32	11	"	29	"	"	"	15,17
P120	2X to 2F1 "	30,33	12	"	13	"	"	"	15,16
P121	2X to 2F1 "	30,12	33	"	13	"	"	"	15,17
P122	2X to 2F2 "	30,6	28	"	5	"	"	"	15,16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P123	2X to 2F2 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	30,38	6	7	5	10	53	ns	15,17
P124	2X to 2F3 "	30,9	8	"	27	"	"	"	15,16
P125	2X to 2F3 "	30,8	9	"	27	"	"	"	15,17
P126	2Y to 2F0 "	32,7,11	Open	30	29	"	"	"	15,16
P127	2Y to 2F0 "	7	11,32	"	29	"	"	"	15,17
P128	2Y to 2F1 "	12,7,33	Open	"	13	"	"	"	15,16
P129	2Y to 2F1 "	7	12,33	"	13	"	"	"	15,17
P130	2Y to 2F2 "	28,7,6	Open	"	5	"	"	"	15,16
P131	2Y to 2F2 "	7	6,28	"	5	"	"	"	15,17
P132	2Y to 2F3 "	8,7,9	Open	"	27	"	"	"	15,16
P133	2Y to 2F3 "	7	8,9	"	27	"	"	"	15,17
P134	3X to 3F0 "	25,3	22	4	21	"	"	"	15,16
P135	3X to 3F0 "	25,22	3	"	21	"	"	"	15,17
P136	3X to 3F1 "	24,23	2	"	24	"	"	"	15,16
P137	3X to 3F1 "	25,2	23	"	24	"	"	"	15,17
P138	3Y to 3F0 "	22,4,3	Open	25	21	"	"	"	15,16
P139	3Y to 3F0 "	4	3,22	25	21	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P140	3Y to 3F1 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	2,4,23	Open	25	24	10	53	ns	15,16
P141	3Y to 3F1 Propagation delay time (t_{PHL}), X and Y address to output, C_L = 150 pF	4	2,23	25	24	10	53	"	15,17
P142	I to $\overline{I}$ Propagation delay time (t_{PHL}), INVERTER C_L = 150 pF	Open	Open	26	31	4	29	"	15,17
Q1	1A0 Propagation delay time (t_{PLH}), A0-A3 to output, C_L = 50 pF	16,37	"	19	18	5	20	"	15,16
Q2	1A1 "	"	"	38	40	"	"	"	"
Q3	1A2 "	"	"	14	35	"	"	"	"
Q4	1A3 "	"	"	15	17	"	"	"	"
Q5	2A0 "	7,30	"	11	29	"	"	"	"
Q6	2A1 "	"	"	33	13	"	"	"	"
Q7	2A2 "	"	"	6	5	"	"	"	"
Q8	2A3 "	"	"	9	27	"	"	"	"
Q9	3A0 "	4,25	"	3	21	"	"	"	"
Q10	3A1 "	4,25	"	23	24	"	"	"	"
Q11	1B0 Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 50 pF	37	16	39	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q12	1B1	Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 50 pF	37	16	20	40	5	20	ns	15,16
Q13	1B2	"	"	"	34	35	"	"	"	"
Q14	1B3	"	"	"	36	17	"	"	"	"
Q15	2B0	"	30	7	32	29	"	"	"	"
Q16	2B1	"	"	"	12	13	"	"	"	"
Q17	2B2	"	"	"	28	5	"	"	"	"
Q18	2B3	"	"	"	8	27	"	"	"	"
Q19	3B0	"	25	4	22	21	"	"	"	"
Q20	3B1	"	25	4	2	24	"	"	"	"
Q21	1B0	Propagation delay time (t_{PLH}), B0-B3 to output (inverting), C_L = 50 pF	16	37	39	18	"	35	"	"
Q22	1B1	"	"	"	20	40	"	"	"	"
Q23	1B2	"	"	"	34	35	"	"	"	"
Q24	1B3	"	"	"	36	17	"	"	"	"
Q25	2B0	"	7	30	32	29	"	"	"	"
Q26	2B1	"	"	"	12	13	"	"	"	15,17
Q27	2B2	"	"	"	28	5	"	"	"	"
Q28	2B3	"	"	"	8	27	"	"	"	"
Q29	3B0	"	4	25	22	21	"	"	"	"
Q30	3B1	"	4	25	2	24	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q31	1X to 1F0 Propagation delay time (t_{PLH}), X and Y address to output, C_L = 50 pF	37,19	39	16	18	5	34	ns	15,16
Q32	1X to 1F0 "	37,39	19	"	18	"	"	"	15,17
Q33	1X to 1F1 "	37,38	20	"	40	"	"	"	15,16
Q34	1X to 1F1 "	37,20	38	"	40	"	"	"	15,17
Q35	1X to 1F2 "	37,14	34	"	35	"	"	"	15,16
Q36	1X to 1F2 "	37,34	14	"	35	"	"	"	15,17
Q37	1X to 1F3 "	37,15	36	"	17	"	"	"	15,16
Q38	1X to 1F3 "	37,36	15	"	17	"	"	"	15,17
Q39	1Y to 1F0 "	39,16,19	Open	37	18	"	"	"	15,16
Q40	1Y to 1F0 "	16	19,39	"	18	"	"	"	15,17
Q41	1Y to 1F1 "	20,16,38	Open	"	40	"	"	"	15,16
Q42	1Y to 1F1 "	16	38,20	"	40	"	"	"	15,17
Q43	1Y to 1F2 "	34,16,14	Open	"	35	"	"	"	15,16
Q44	1Y to 1F2 "	16	14,34	"	35	"	"	"	15,17
Q45	1Y to 1F3 "	36,16,15	Open	"	17	"	"	"	15,16
Q46	1Y to 1F3 "	16	15,36	"	17	"	"	"	15,17
Q47	2X to 2F0 "	30,11	32	7	29	"	"	"	15,16
Q48	2X to 2F0 "	30,32	11	"	29	"	"	"	15,17
Q49	2X to 2F1 "	30,33	12	"	13	"	"	"	15,16

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q50	2X to 2F1	Propagation delay time (t_{PLH}), X and Y address to output, C_L = 50 pF	30,12	33	7	13	5	34	ns	15,17
Q51	2X to 2F2	"	30,6	28	"	5	"	"	"	15,16
Q52	2X to 2F2	"	30,48	6	"	5	"	"	"	15,17
Q53	2X to 2F3	"	30,9	8	"	27	"	"	"	15,16
Q54	2X to 2F3	"	30,8	9	"	27	"	"	"	15,17
Q55	2Y to 2F0	"	32,7,11	Open	30	29	"	"	"	15,16
Q56	2Y to 2F0	"	7	11,32	"	29	"	"	"	15,17
Q57	2Y to 2F1	"	12,7,33	Open	"	13	"	"	"	15,16
Q58	2Y to 2F1	"	7	33,12	"	13	"	"	"	15,17
Q59	2Y to 2F2	"	28,7,6	Open	"	5	"	"	"	15,16
Q60	2Y to 2F2	"	7	6,28	"	5	"	"	"	15,17
Q61	2Y to 2F3	"	8,7,9	Open	"	27	"	"	"	15,16
Q62	2Y to 2F3	"	7	8,9	"	27	"	"	"	15,17
Q63	3X to 3F0	"	25,3	22	4	21	"	"	"	15,16
Q64	3X to 3F0	"	24,22	3	"	21	"	"	"	15,17
Q65	3X to 3F1	"	25,23	2	"	24	"	"	"	15,16
Q66	3X to 3F1	"	25,2	23	"	24	"	"	"	15,17
Q67	3Y to 3F0	"	22,4,3	Open	25	21	"	"	"	15,16
Q68	3Y to 3F0	"	4	3,22	25	21	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q69	3Y to 3F1 Propagation delay time (t_{PLH}), X and Y address to output, C_L = 50 pF	2,4,23	Open	25	24	5	34	ns	15,16
Q70	3Y to 3F1 Propagation delay time (t_{PLH}), X and Y address to output, C_L = 50 pF	4	2,23	25	24	5	34	"	15,17
Q71	1 to $\overline{1}$ Propagation delay time (t_{PLH}), INVERTER, C_L = 50 pF	Open	Open	26	31	3	19	"	15,17
Q72	1A0 Propagation delay time (t_{PLH}), A0-A3 to output, C_L = 150 pF	16,37	"	19	18	5	25	"	15,16
Q73	1A1 "	"	"	38	40	"	"	"	"
Q74	1A2 "	"	"	14	35	"	"	"	"
Q75	1A3 "	"	"	15	17	"	"	"	"
Q76	2A0 "	7,30	"	11	29	"	"	"	"
Q77	2A1 "	"	"	33	13	"	"	"	"
Q78	2A2 "	"	"	6	5	"	"	"	"
Q79	2A3 "	"	"	9	27	"	"	"	"
Q80	3A0 "	4,25	"	3	21	"	"	"	"
Q81	3A1 "	4,25	"	23	24	"	"	"	"
Q82	1B0 Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 150 pF	37	16	39	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026[illegible]-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q83	1B1	Propagation delay time (t_{PLH}), B0-B3 to output (noninverting), C_L = 150 pF	37	16	20	40	5	25	ns	15,16
Q84	1B2	"	"	"	34	35	"	"	"	"
Q85	1B3	"	"	"	36	17	"	"	"	"
Q86	2B0	"	30	7	32	29	"	"	"	"
Q87	2B1	"	"	"	12	13	"	"	"	"
Q88	2B2	"	"	"	28	5	"	"	"	"
Q89	2B3	"	"	"	8	27	"	"	"	"
Q90	3B0	"	25	4	22	21	"	"	"	"
Q91	3B1	"	25	4	2	24	"	"	"	"
Q92	1B0	Propagation delay time (t_{PLH}), B0-B3 to output (inverting), C_L = 150 pF	16	37	39	18	"	40	"	15,17
Q93	1B1	"	"	"	20	40	"	"	"	"
Q94	1B2	"	"	"	34	35	"	"	"	"
Q95	1B3	"	"	"	36	17	"	"	"	"
Q96	2B0	"	7	30	32	29	"	"	"	"
Q97	2B1	"	"	"	12	13	"	"	"	"
Q98	2B2	"	"	"	28	5	"	"	"	"
Q99	2B3	"	"	"	8	27	"	"	"	"
Q100	3B0	"	4	25	22	21	"	"	"	"
Q101	3B1	"	4	25	2	24	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q102	1X to 1F0 Propagation delay time (t_{PLH}), X and Y address to output, C_L = 150 pF	37,19	39	16	18	10	39	ns	15,16
Q103	1X to 1F0 "	37,39	19	"	18	"	"	"	15,17
Q104	1X to 1F1 "	37,38	20	"	40	"	"	"	15,16
Q105	1X to 1F1 "	37,20	38	"	40	"	"	"	15,17
Q106	1X to 1F2 "	37,14	34	"	35	"	"	"	15,16
Q107	1X to 1F2 "	37,34	14	"	35	"	"	"	15,17
Q108	1X to 1F3 "	37,15	36	"	17	"	"	"	15,16
Q109	1X to 1F3 "	37,36	15	"	17	"	"	"	15,17
Q110	1Y to 1F0 "	39,16,19	Open	37	18	"	"	"	15,16
Q111	1Y to 1F0 "	16	19,39	"	18	"	"	"	15,17
Q112	1Y to 1F1 "	20,16,38	Open	"	40	"	"	"	15,16
Q113	1Y to 1F1 "	16	20,38	"	40	"	"	"	15,17
Q114	1Y to 1F2 "	34,16,14	Open	"	35	"	"	"	15,16
Q115	1Y to 1F2 "	16	14,34	"	35	"	"	"	15,17
Q116	1Y to 1F3 "	36,16,15	Open	27	17	"	"	"	15,16
Q117	1Y to 1F3 "	16	15,36	37	17	"	"	"	15,17
Q118	2X to 2F0 "	30,11	32	7	29	"	"	"	15,16
Q119	2X to 2F0 "	30,32	11	"	29	"	"	"	15,17
Q120	2X to 2F1 "	30,33	12	"	13	"	"	"	15,16
Q121	2X to 2F1 "	30,12	33	"	13	"	"	"	15,17

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

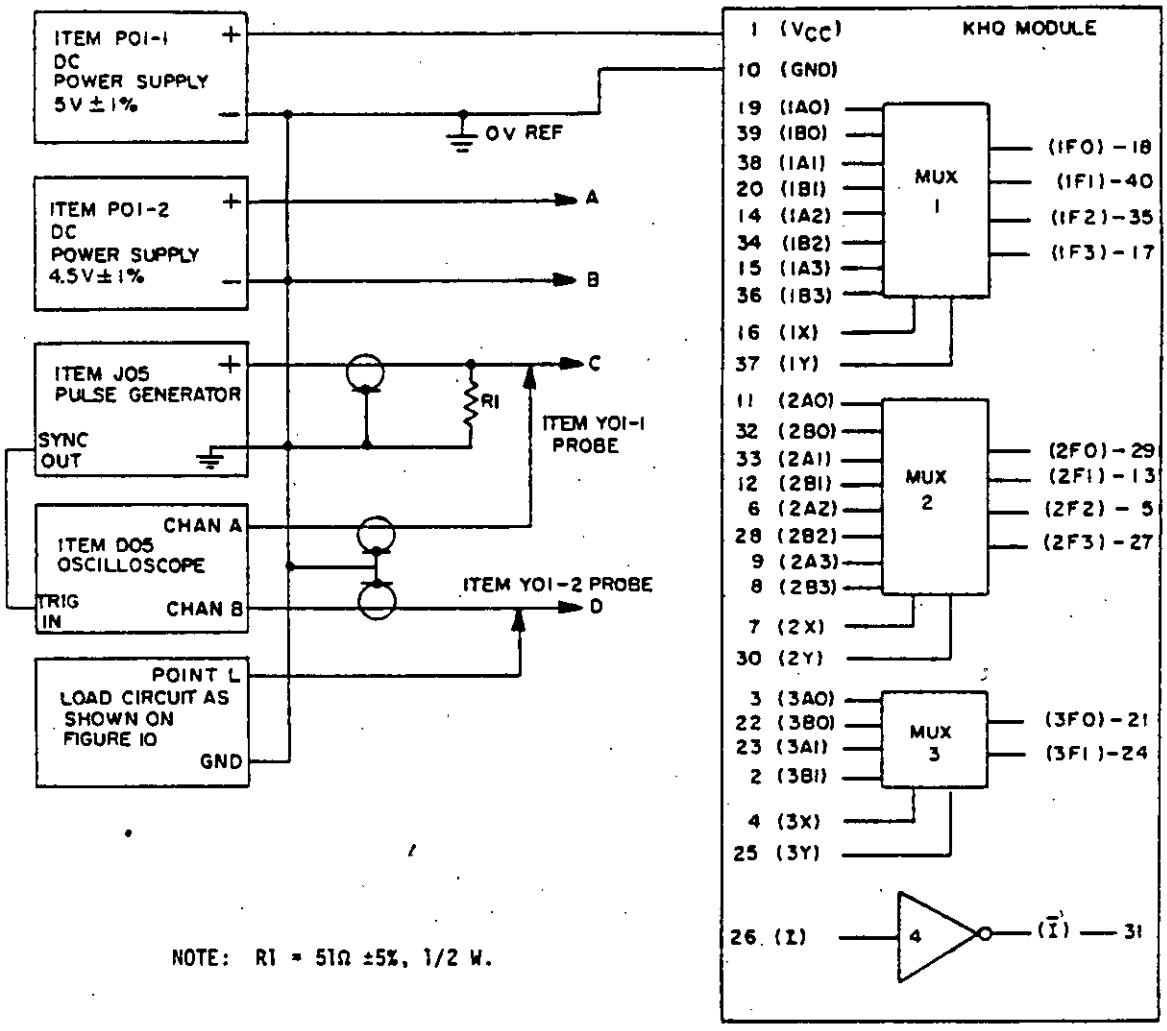
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q122	2X to 2F2	Propagation delay time (t_{PLH}), X and Y address to output, C_L = 150 pF	30,6	28	7	5	10	39	ns	15,16
Q123	2X to 2F2	"	30,28	6	"	5	"	"	"	15,17
Q124	2X to 2F3	"	30,9	8	"	27	"	"	"	15,16
Q125	2X to 2F3	"	30,8	9	"	27	"	"	"	15,17
Q126	2Y to 2F0	"	32,7,11	Open	30	29	"	"	"	15,16
Q127	2Y to 2F0	"	7	11,32	"	29	"	"	"	15,17
Q128	2Y to 2F1	"	12,7,33	Open	"	13	"	"	"	15,16
Q129	2Y to 2F1	"	7	12,33	"	13	"	"	"	15,17
Q130	2Y to 2F2	"	28,7,6	Open	"	5	"	"	"	15,16
Q131	2Y to 2F2	"	7	6,28	"	5	"	"	"	15,17
Q132	2Y to 2F3	"	8,7,9	Open	"	27	"	"	"	15,16
Q133	2Y to 2F3	"	7	8,9	"	27	"	"	"	15,17
Q134	3X to 3F0	"	25,3	22	4	21	"	"	"	15,16
Q135	3X to 3F0	"	25,22	3	"	21	"	"	"	15,17
Q136	3X to 3F1	"	25,23	2	"	24	"	"	"	15,16
Q137	3X to 3F1	"	25,2	23	"	24	"	"	"	15,17
Q138	3Y to 3F0	"	22,4,3	Open	25	21	"	"	"	15,16
Q139	3Y to 3F0	"	4	3,22	"	21	"	"	"	15,17
Q140	3Y to 3F1	"	2,4,23	Open	"	24	"	"	"	15,16

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q141	3Y to 3F1 Propagataion delay time (t_{PLH}), X and Y address to output, C_L = 150 pF	4	2,23	25	24	10	39	ns	15,17
Q142	I to $\overline{I}$ Propagation delay time (t_{PLH}), INVERTER, C_L = 150 pF	Open	Open	26	31	4	29	ns	15,17

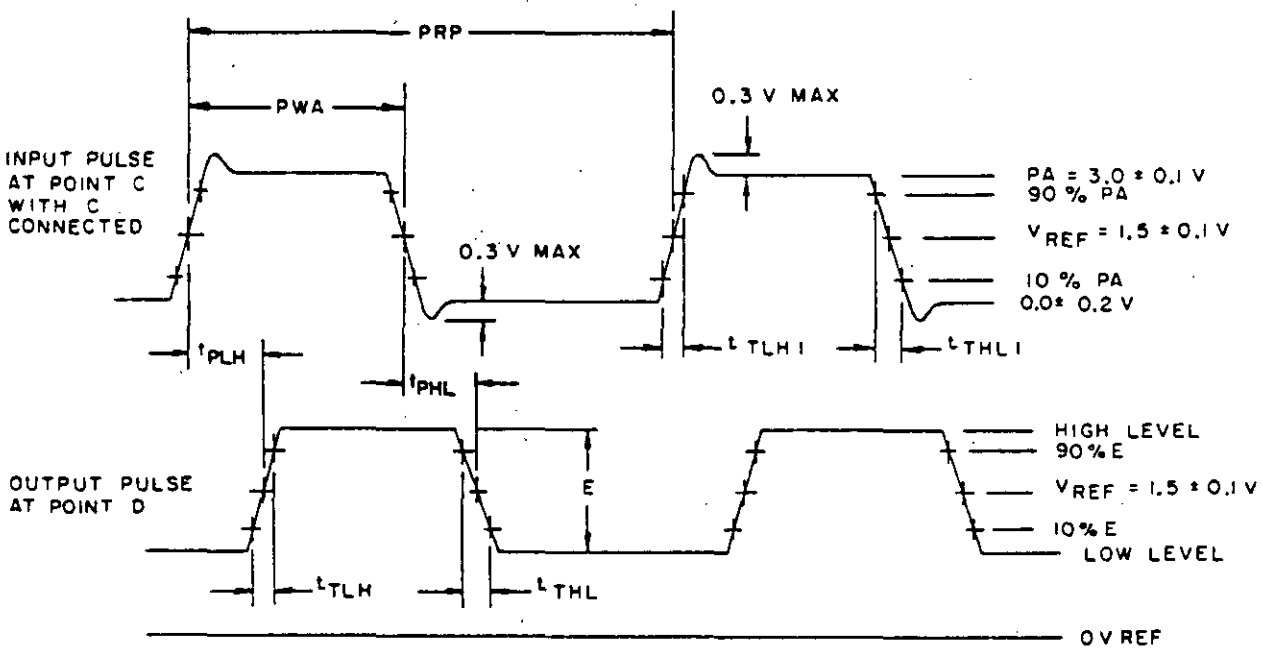
Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:53Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIV and XV.

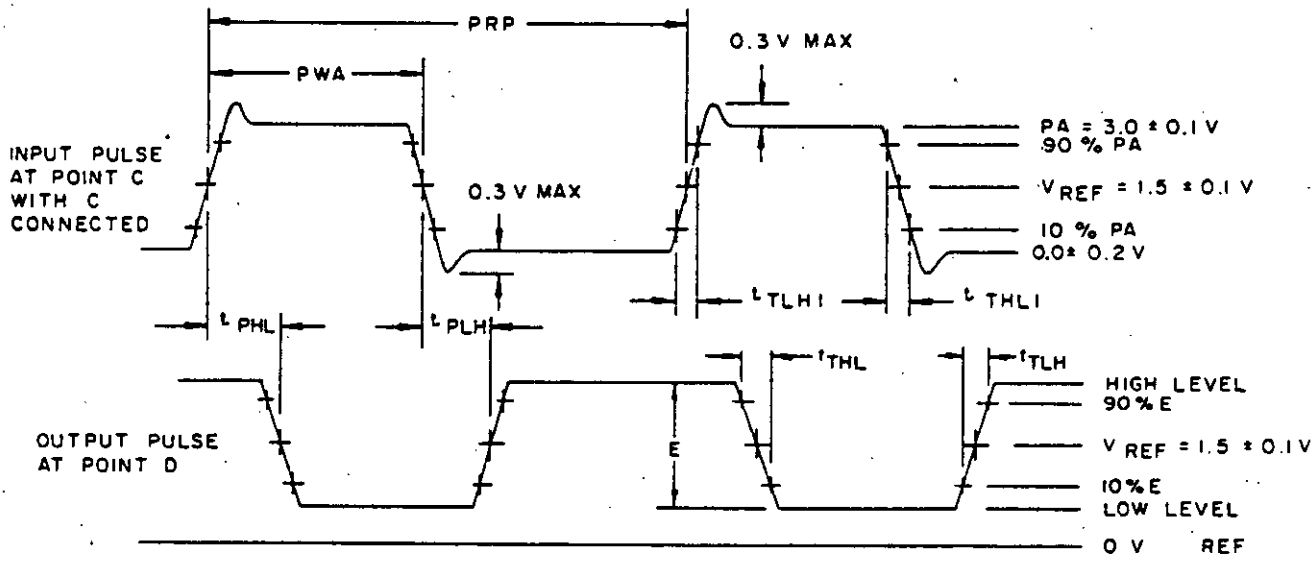
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items YO1-1 and YO1-2 and output pins; and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIV and XV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 500 ±10 ns.
   PRP = 1,000 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between items YO1-1 and YO1-2 and input and output pins; and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIV and XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level) test.

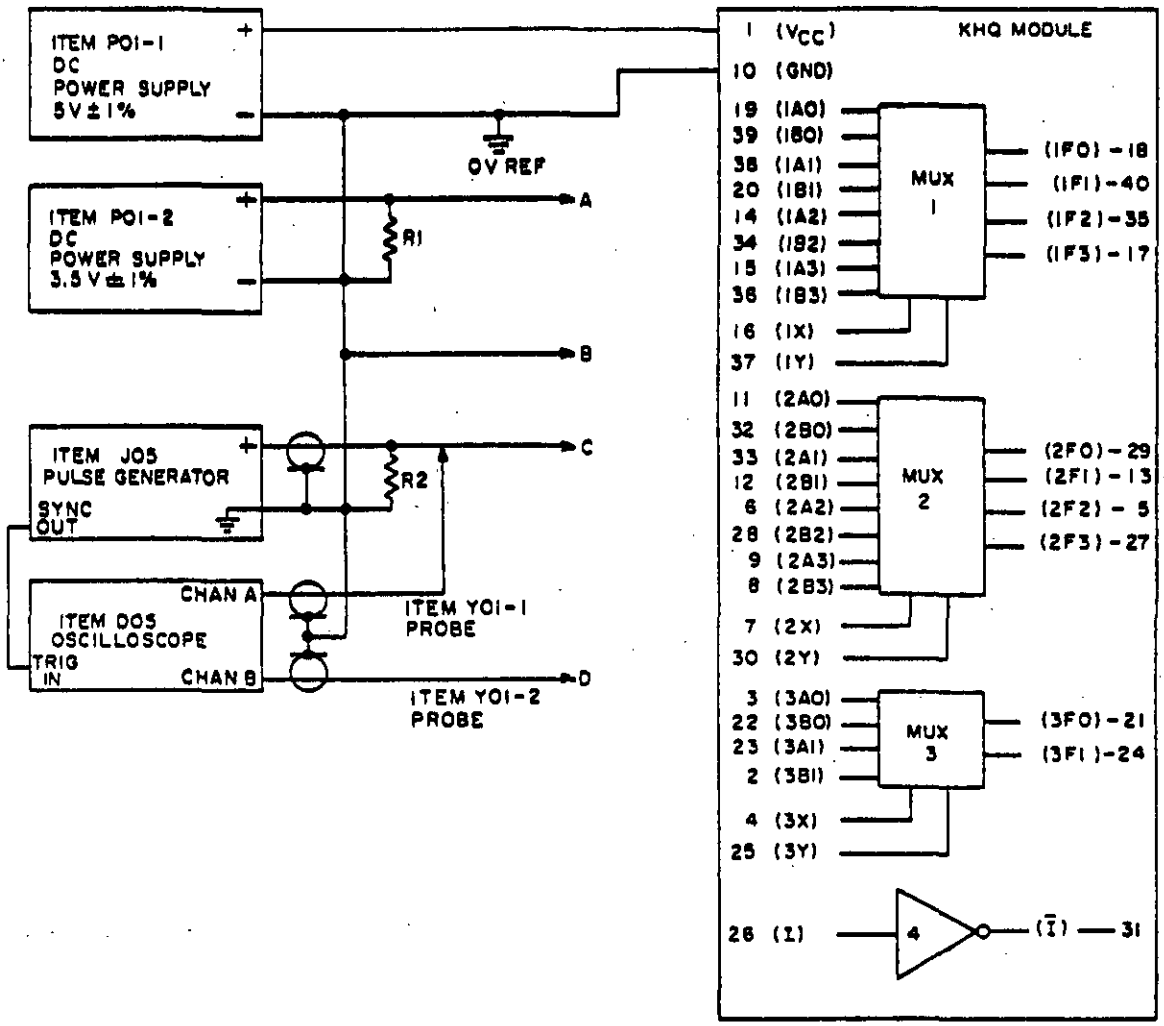
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1F0	Output crosstalk, low level	16,7,4,39	37,30,25	20,34,36, 32,12,28, 8,22,2,26	18	0	600	mV	18,19
R2	1F1	"	16,7,4,20	"	39,34,36, 32,12,28, 8,22,2,26	40	"	"	"	"
R3	1F2	"	16,7,4,34	"	39,20,36, 32,12,28, 8,22,2,26	35	"	"	"	"
R4	1F3	"	16,7,4,36	"	39,20,34, 32,12,28, 8,22,2,26	17	"	"	"	"
R5	2F0	"	16,7,4,32	"	39,20,34, 36,12,28, 8,22,2,26	29	"	"	"	"
R6	2F1	"	16,7,4,12	"	39,20,34, 36,28,32, 8,22,2,26	13	"	"	"	"
R7	2F2	"	16,7,4,28	"	39,20,34, 36,32,12, 8,22,2,26	5	"	"	"	"
R8	2F3	"	16,7,4,8	"	39,20,34, 36,32,12, 28,22,2,26	27	"	"	"	"
R9	3F0	"	16,7,4,22	"	39,20,34, 36,32,12, 28,8,2,26	21	"	"	"	"
R10	3F1	"	16,7,4,2	"	39,20,34, 36,32,12, 28,8,22,26	24	"	"	"	"
R11	$\overline{I}$	"	16,7,4,26	"	39,20,34, 36,32,12, 28,8,22,2	31	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260[illegible]-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1F0	Output crosstalk, high level	16,7,4	37,30, 25,39	20,34,36, 32,12,28, 8,22,2,26	18	0	400	mV	18,19
S2	1F1	"	"	37,30, 25,20	39,34,36, 32,12,28, 8,22,2,26	40	"	"	"	"
S3	1F2	"	"	37,30, 25,34	39,20,36, 32,12,28, 8,22,2,26	35	"	"	"	"
S4	1F3	"	"	37,30, 25,36	39,20,34, 32,12,28, 8,22,2,26	17	"	"	"	"
S5	2F0	"	"	37,30, 25,32	39,20,34, 36,12,28, 8,22,2,26	29	"	"	"	"
S6	2F1	"	"	37,30, 25,12	39,20,34, 36,32,28, 8,22,2,26	13	"	"	"	"
S7	2F2	"	"	37,30, 25,28	39,20,34, 36,32,12, 8,22,2,26	5	"	"	"	"
S8	2F3	"	"	37,30, 25,8	39,20,34, 36,32,12, 28,22,2,26	27	"	"	"	"
S9	3F0	"	"	37,30, 25,22	39,20,34, 36,32,12, 28,8,2,26	21	"	"	"	"
S10	3F1	"	"	37,30, 25,2	39,20,34, 36,32,12, 28,8,22,26	24	"	"	"	"
S11	$\overline{I}$	"	"	37,30, 25,26	39,20,34, 36,32,12, 28,8,22,2	31	"	"	"	"

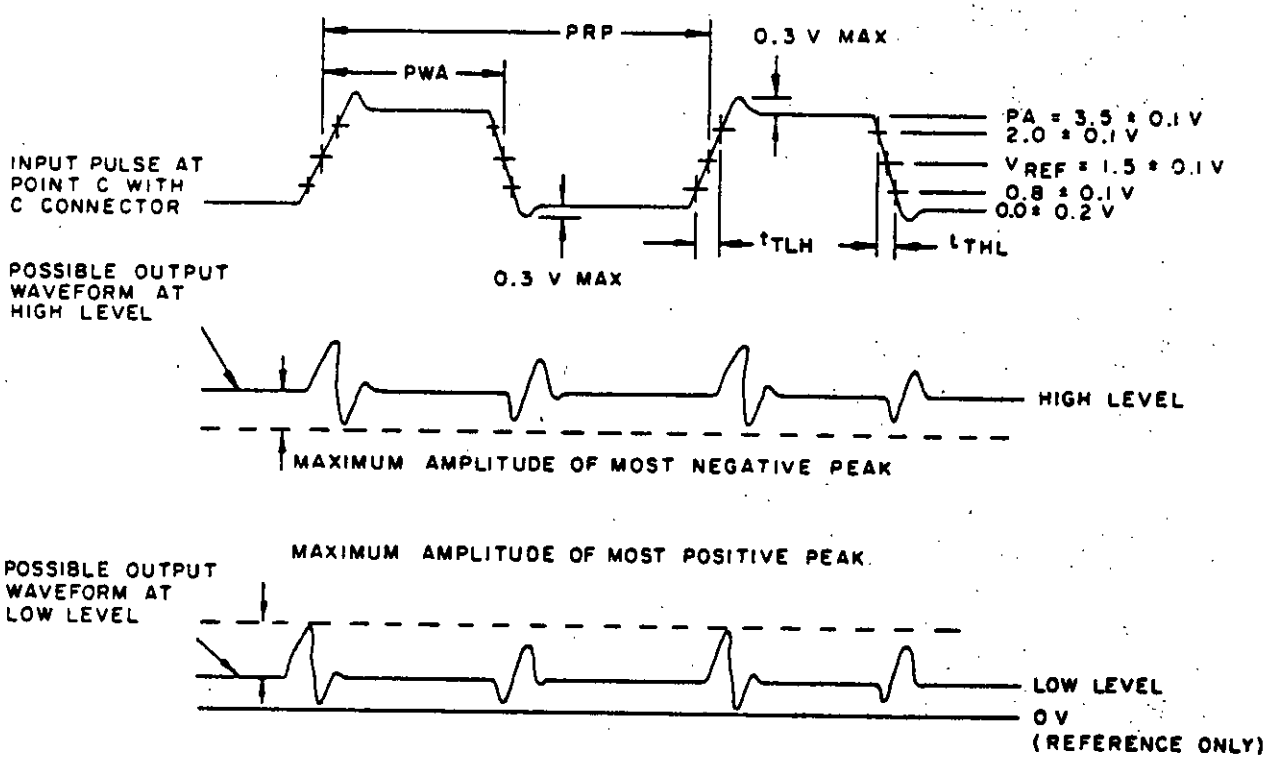
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 10 with the following exception: capacitor C_L used in KHQ load circuit is 50 pF for test tables XVI and XVII.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at module pins)
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R2 and input pins; between items YO1-1 and YO1-2 and input and output pins; between load circuits and output pins; and between ground leads of items YO1-1 and YO1-2 and test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.6	3.0	µF	22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test.

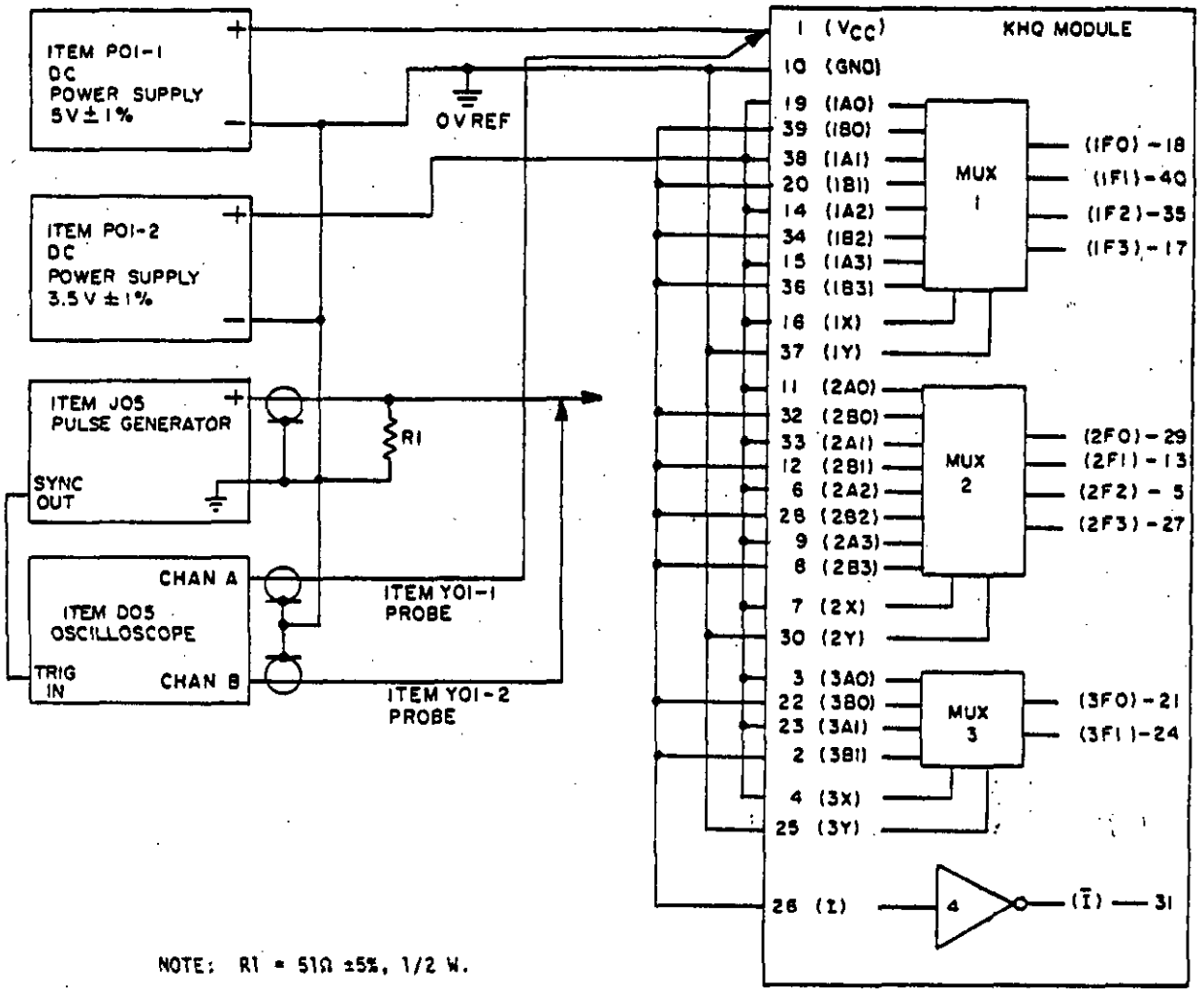
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1A0 Input capacitance	19	Open	2	15	pF	22
V2	1A1 "	38	"	"	"	"	"
V3	1A2 "	14	"	"	"	"	"
V4	1A3 "	15	"	"	"	"	"
V5	2A0 "	11	"	"	"	"	"
V6	2A1 "	33	"	"	"	"	"
V7	2A2 "	6	"	"	"	"	"
V8	2A3 "	9	"	"	"	"	"
V9	3A0 "	3	"	"	"	"	"
V10	3A1 "	23	"	"	"	"	"
V11	3X "	4	"	"	"	"	"
V12	3Y "	25	"	"	"	"	"
V13	I "	26	"	"	20	"	"
V14	1B0 "	39	"	"	30	"	"
V15	1B1 "	20	"	"	"	"	"
V16	1B2 "	34	"	"	"	"	"
V17	1B3 "	36	"	"	"	"	"
V18	1X "	16	"	"	"	"	"
V19	1Y "	37	"	"	"	"	"
V20	2B0 "	32	"	"	"	"	"
V21	2B1 "	12	"	"	"	"	"
V22	2B2 "	28	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V23	2B3 Input capacitance	8	Open	2	30	pF	22
V24	2X "	7	"	"	"	"	"
V25	2Y "	30	"	"	"	"	"
V26	3B0 "	22	"	"	"	"	"
V27	3B1 "	2	"	"	"	"	"

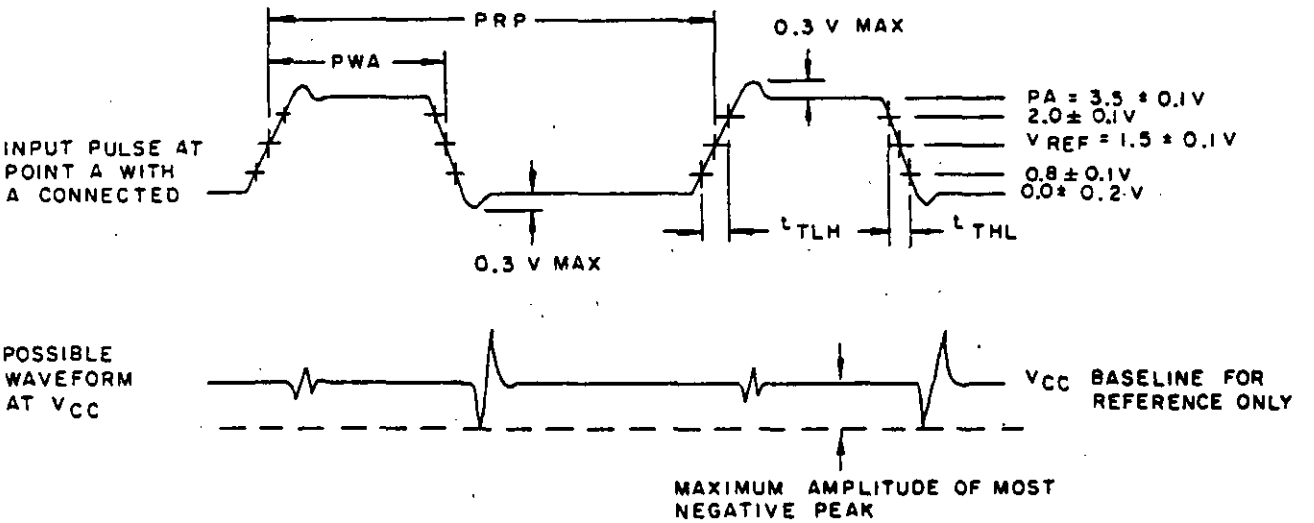
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVIII.

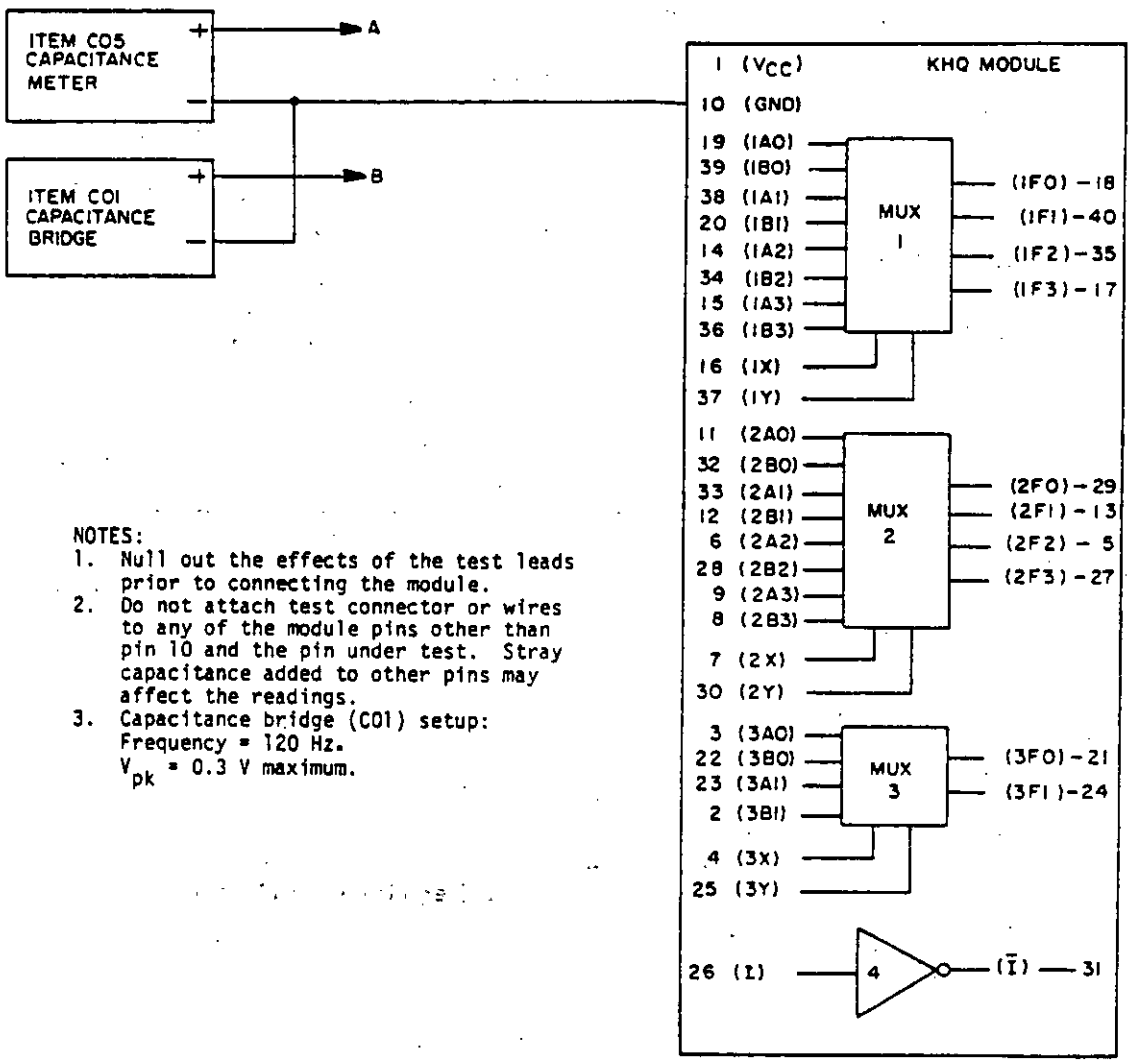
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at module pins).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:53Z
Check the source to verify that this is the currentversion before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup: Frequency = 120 Hz. V_{pk} = 0.3 V maximum.

FIGURE 22. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/72B(EC); Modules, Standard Electronic Multiplexer, Digital, Key Code KHQ

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:53Z
Check the source to verify that this is the current version before use.